Vector Autoregressive Models (VARs)

Econometrics. Lecture 13

Content

1. Introduction to VARs
2. Stationarity and lag specification criteria
3. Residual diagnostics
4. Forecasting with VARs
5. SVARs
6. Impulse responses & variance decomposition

1. Introduction to VARs

Where VARs came from

Large-scale macro-econometric models of the 1960s and 1970s achieved identification by declaring most variables exogenous and most cross-equation channels absent. Sims (1980) argued these restrictions were asserted rather than tested, and that they were not credible.

Proposed VARs as an alternative: let every variable in the system depend on lags of every other, estimate that, and impose theory only at the point where you need it.

Sims, C. A. (1980). Macroeconomics and Reality. *Econometrica*, *48*(1), 1–48. https://doi.org/10.2307/1912017

What a VAR is

- A system of *n* equations in *n* variables, where each equation regresses one variable on *p* lags of every variable in the system, including its own.

- Every variable is endogenous.

- Because every equation has an identical set of regressors, the system needs no more machinery than OLS run *n* times.

What are VARs used for?

	Reduced-form VAR	Structural VAR
Question	What comes next?	What caused what?
Needs	Estimation only	Estimation and identification
Output	Forecasts, scenarios	Impulse responses, variance decompositions

The smallest case: two variables, one lag

$$y_{1,t} = a_{10} + a_{11}y_{1,t-1} + a_{12}y_{2,t-1} + e_{1,t},$$
$$y_{2,t} = a_{20} + a_{21}y_{1,t-1} + a_{22}y_{2,t-1} + e_{2,t}$$

Matrix form:

$y_t = a_0 + A_1 y_{t-1} + e_t,$ where

(vector with the value of variables at time t)

$y_t = [y_{1,t}, y_{2,t}]'$

(vector of constants)

$a_0 = [a_{10}, a_{20}]'$

(matrix of coefficients)

$A_1 = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

(vector of errors)

$e_t = [e_{1,t}, e_{2,t}]'$

Error assumptions:

$\mathrm{E}[\mathrm{e}_t] = 0$

$\mathrm{E}[e_t e_t'] = \Sigma_e = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$

$\mathrm{E}[e_t e_s] = 0$ for $t \neq s$

The general case. VAR(*p*)

Stack the *n* variables into y_t and the model is a matrix AR(*p*):

$$y_t = [y_{1,t}, y_{2,t}, \cdots, y_{n,t}]'$$

$$y_t = a_0 + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + e_t$$

Reduced-form VAR

a_0 – *(n x 1)* vector of intercepts

A_{i} – *(n x n)* matrices of coefficients, *i = 1, …, p*

e_t – *(n x 1)* vector of reduced-form errors

$$\mathrm{E}[e_t] = 0$$

$$\mathrm{E}[e_t e_t'] = \Sigma_e$$

$$\mathrm{E}[e_t e_s'] = 0 \text{ for } t \neq s$$

Σ_e is symmetric and positive definite, but generally not diagonal.

Estimation and specification choices

Estimation:

- Run OLS equation by equation.
- Estimators are **consistent** and asymptotically **efficient** provided the lag order is right, …
- … if it is not, the residuals are autocorrelated and the estimators are consistent for a model nobody wanted.

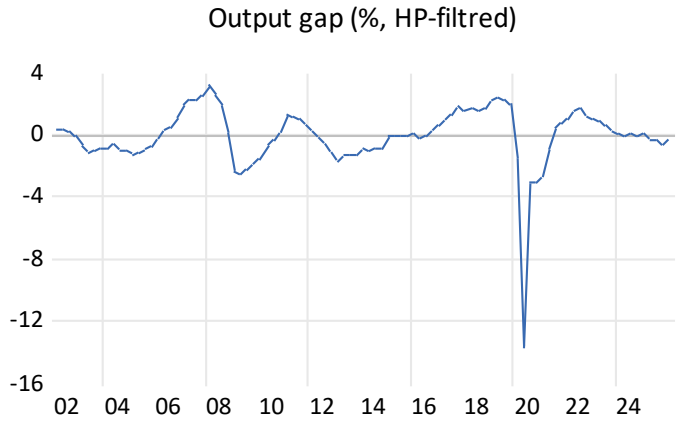
Specification:

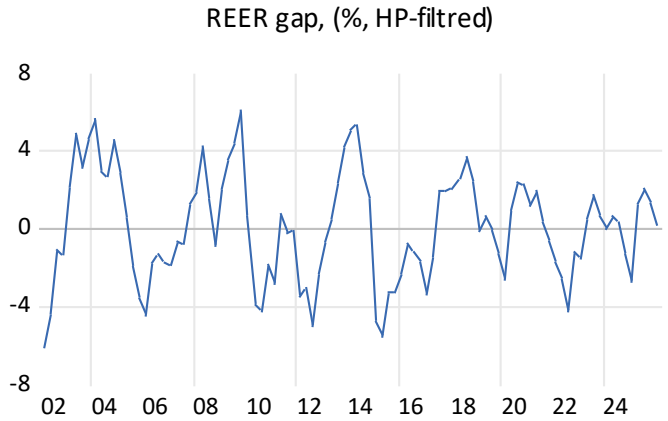
- **Which variables:** driven by the question, by theory, and by what the sample will bear.
- **Which transformation:** levels, gaps, log differences or growth rates.
- **What else on the right-hand side:** constant, time trend, dummies, genuinely exogenous series.

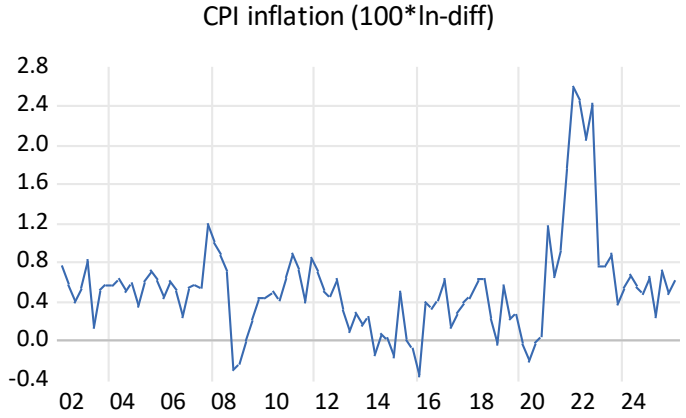
EViews example. EA VAR

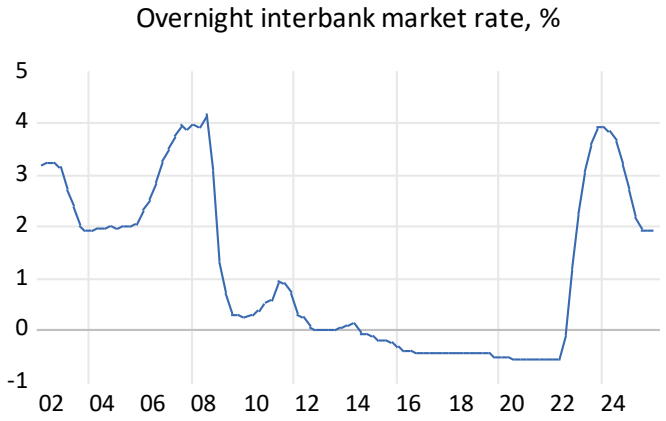
Quarterly euro area data, 2002Q1 to 2026Q1.

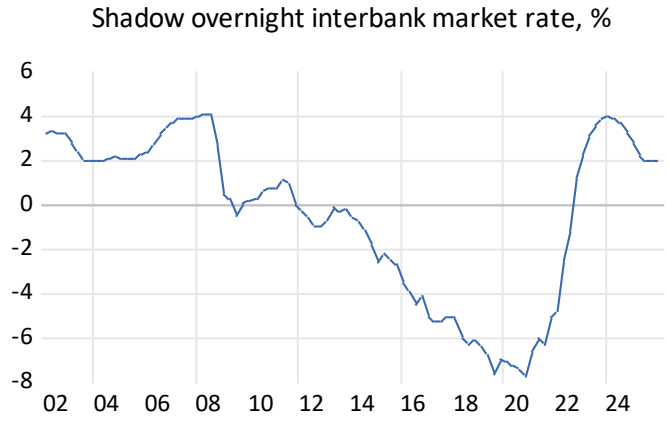
Four endogenous variables – the output gap (from HP-filtered log GDP), CPI inflation, the monetary policy rate (actual and shadow), and the real effective exchange rate gap (HP-filtered).





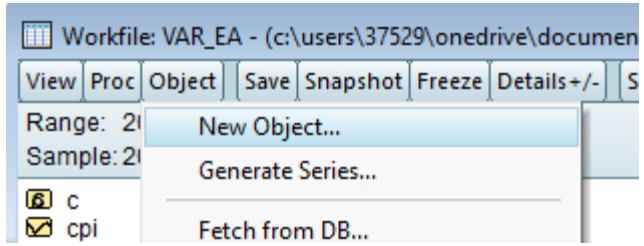


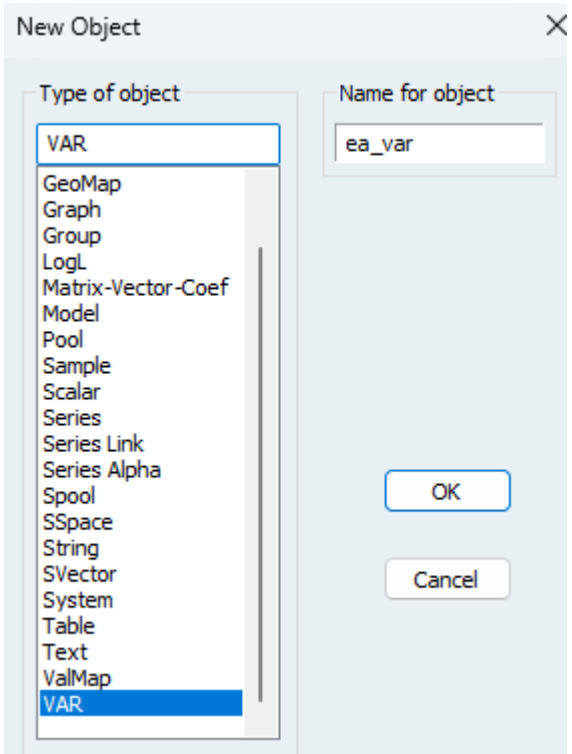




EViews example. Estimation of a VAR (1)

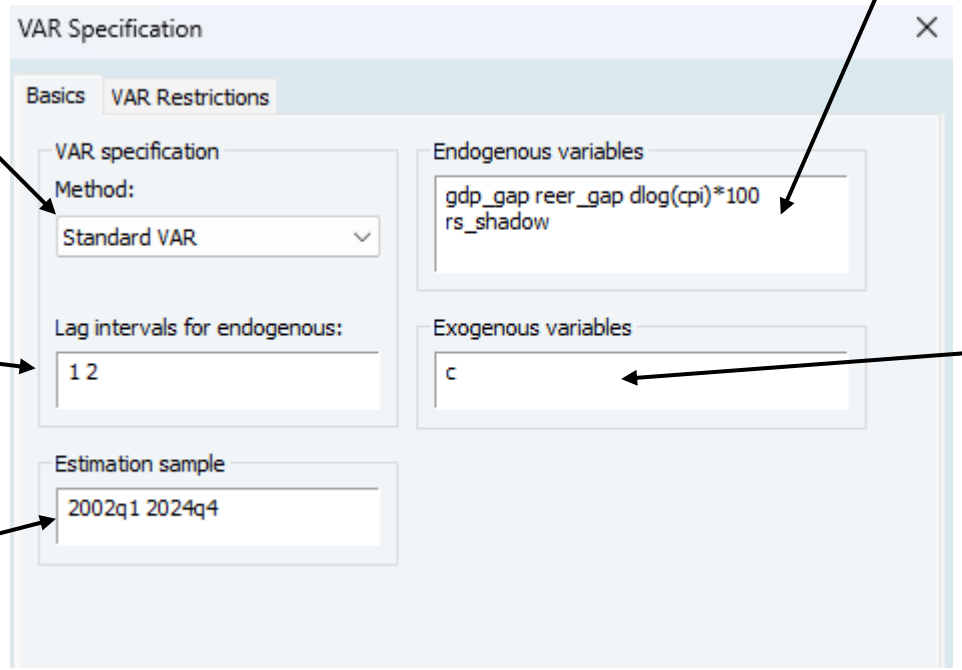
Create a VAR object





Choose the method

Select endogenous variables



Specify the lag length

Adjust the estimation sample

Add exogenous variables if necessary

EViews example. Estimation of a VAR (2)

Vector Autoregression Estimates
Date: 07/30/26 Time: 12:45
Sample: 2002Q1 2024Q4
Included observations: 92
Standard errors in ( ) & t-statistics in [ ]

Individual equations

	GDP_GAP	REER_GAP	DLOG(CPI)*...	RS_SHADOW
GDP_GAP(-1)	0.530275	-0.062709	0.005431	-0.027848
	(0.11150)	(0.12580)	(0.02444)	(0.03261)
	[4.75603]	[-0.49847]	[0.22218]	[-0.85389]
GDP_GAP(-2)	0.019481	0.024038	-0.026653	-0.054194
	(0.11150)	(0.12580)	(0.02444)	(0.03261)
	[0.17473]	[0.19107]	[-1.09033]	[-1.66169]
REER_GAP(-1)	0.114962	0.964328	-0.025860	-0.013498
	(0.09418)	(0.10627)	(0.02065)	(0.02755)
	[1.22064]	[9.07441]	[-1.25239]	[-0.48996]
REER_GAP(-2)	-0.111320	-0.296594	0.024798	-0.009339
	(0.09343)	(0.10542)	(0.02048)	(0.02733)
	[-1.19144]	[-2.81336]	[1.21058]	[-0.34170]
DLOG(CPI(-1))*100	0.828446	-0.333793	0.711454	0.595960
	(0.53103)	(0.59919)	(0.11643)	(0.15533)
	[1.56006]	[-0.55708]	[6.11079]	[3.83664]
DLOG(CPI(-2))*100	-0.193518	0.293152	0.148823	0.273710
	(0.53386)	(0.60238)	(0.11705)	(0.15616)
	[-0.36249]	[0.48666]	[1.27150]	[1.75275]
RS_SHADOW(-1)	-0.032320	-0.336681	-0.081126	1.001640
	(0.39044)	(0.44055)	(0.08560)	(0.11421)
	[-0.08278]	[-0.76423]	[-0.94772]	[8.77033]
RS_SHADOW(-2)	0.043445	0.352913	0.069506	-0.028132
	(0.38567)	(0.43517)	(0.08456)	(0.11281)
	[0.11265]	[0.81097]	[0.82200]	[-0.24937]
C	-0.338436	0.060566	0.068302	-0.473504
	(0.33104)	(0.37352)	(0.07258)	(0.09683)
	[-1.02236]	[0.16215]	[0.94109]	[-4.88997]
R-squared	0.385558	0.619686	0.582944	0.984950
Adj. R-squared	0.326334	0.583030	0.542746	0.983499
Sum sq. resids	220.2632	280.4276	10.58751	18.84640
S.E. equation	1.629041	1.838109	0.357156	0.476513
F-statistic	6.510229	16.90512	14.50176	678.9733
Log likelihood	-170.7019	-181.8106	-31.08511	-57.61087
Akaike AIC	3.906564	4.148056	0.871416	1.448062
Schwarz SC	4.153261	4.394753	1.118112	1.694759
Mean dependent	-0.007854	0.077110	0.525227	-0.614298
S.D. dependent	1.984768	2.846550	0.528177	3.709529

Determinant resid covariance (dof adj.)	0.205710
Determinant resid covariance	0.136275
Log likelihood	-430.4876
Akaike information criterion	10.14104
Schwarz criterion	11.12782
Number of coefficients	36

Coefficients and their (standard errors) and [*t*-stats]

Constants

Usual statistics

The R^2 values are high, and that is normal for VARs – many lags means a lot of information, and a good in-sample fit.

2. Stationarity and lag specification criteria

What stationarity means for a system

A VAR is **covariance-stationary** when:

1. The mean vector $\mu = \mathrm{E}[y_t]$ is the same at every t:

$$\mathrm{E}[y_t] = \mathrm{E}[y_{t+j}] = \mu = (\mu_1, \mu_2, \ldots, \mu_n)'$$

2. The autocovariance matrices Γ_y depend on the distance h only, not on where in the sample you are standing:

$$\mathrm{E}[(y_t - \mu)(y_{t+h} - \mu)'] = \mathrm{E}[(y_s - \mu)(y_{s+h} - \mu)'] = \Gamma_y(h)$$

Every stationary VAR is an infinite sum of past shocks

Invert the lag polynomial :

$$y_t = \mu + e_t + \Psi_1 \mathrm{e}_{\mathrm{t}-1} + \Psi_2 \mathrm{e}_{\mathrm{t}-2} + \cdots = \mu + \sum_{i=0}^{\infty} \Psi_i e_{t-i}\,,$$

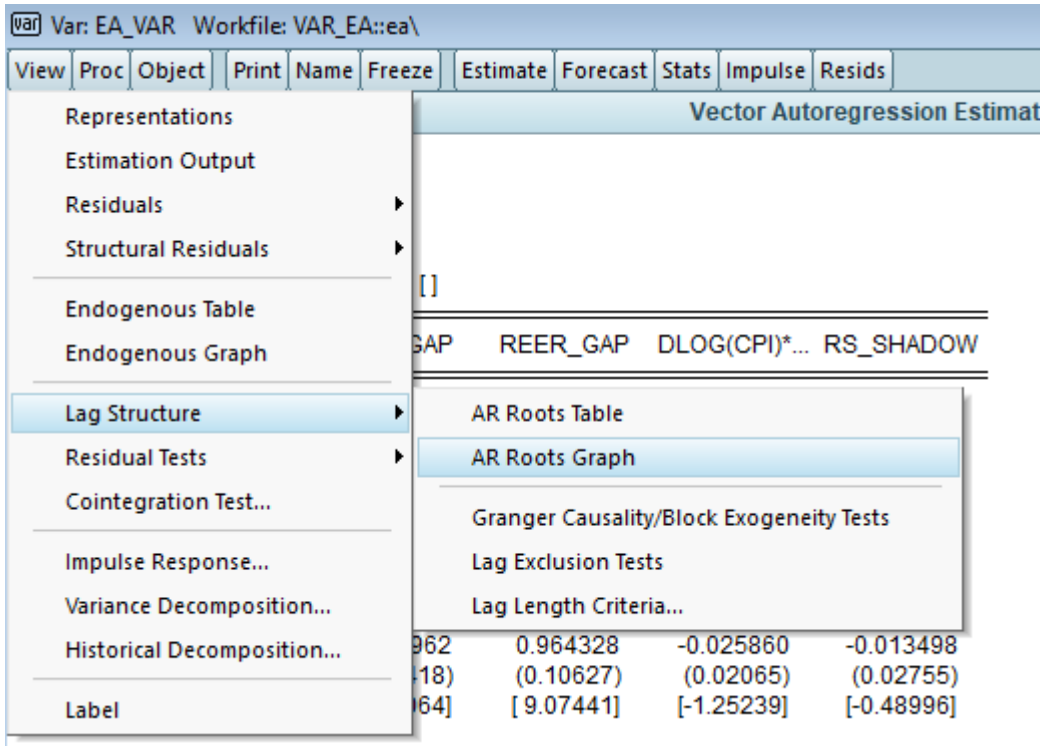
where
Ψ_i is a (n x n) matrix of coefficients.

Read the Ψ_i as coefficients: Ψ_i is how much a one-unit reduced-form error i periods ago still moves y_t today.

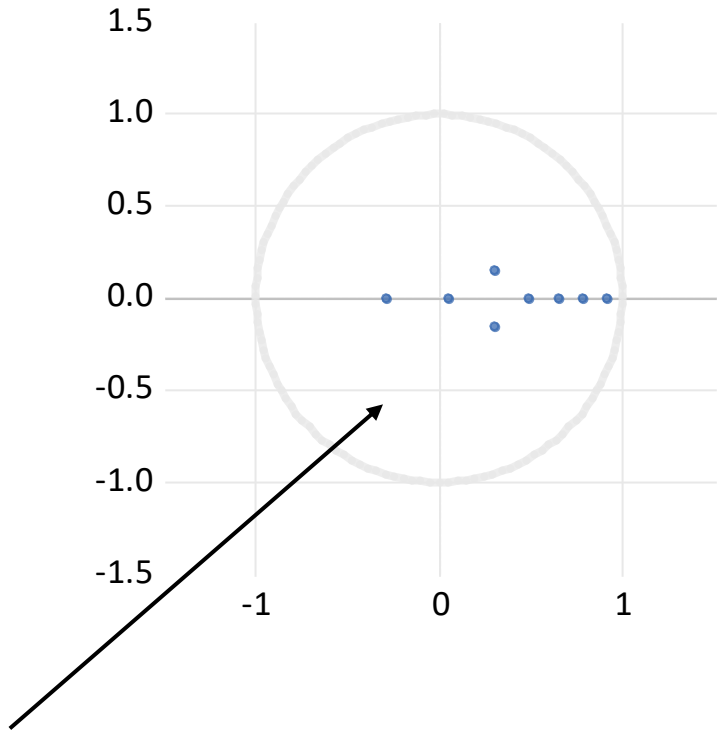
Stationarity is what makes this converge. Without it the sum does not exist and there is nothing to plot.

These are not yet impulse responses. Ψ_i traces reduced-form errors, which are correlated with each other.

Checking stationarity in EViews



Inverse Roots of AR Characteristic Polynomial



All inverse roots of the characteristic AR polynomial lie inside the unit circle – **VAR is stationary**

Lags needed for the VAR

What number of lags is most appropriate?

1. **Too few.** If p is extremely short, the model may be poorly specified.

2. **Too many.** If p is extremely long, too many degrees of freedom will be lost.

3. **Enough.** p should be large enough that the residuals behave as individual white noises.

Over-fitting vs. omitted variable bias

Over-fitting problem

Too many coefficients chase too little data → poor-quality estimates and bad out-of-sample forecasts, even with a great in-sample fit.

Omitted-variable bias

Trimming variables or lags to save degrees of freedom can leave out important variables or dynamics – biasing the estimators.

Possible solutions

1. **Core VAR + rotating variables**
2. **Bayesian analysis**

Choosing p

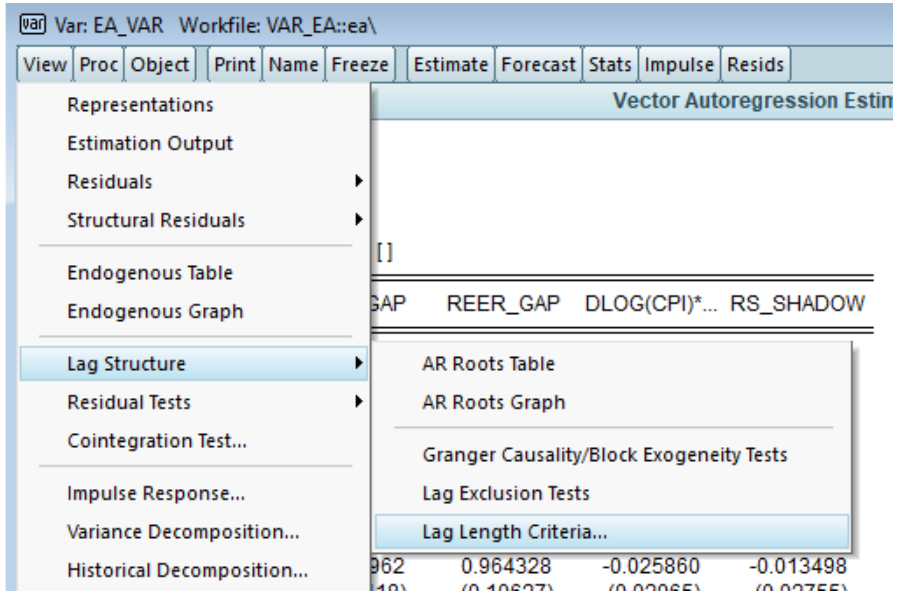
AIC, SC (Schwarz) and HQ (Hannan–Quinn) each penalise added parameters differently. SC and HQ are consistent for the true order; AIC is not, and tends to over-select in finite samples.

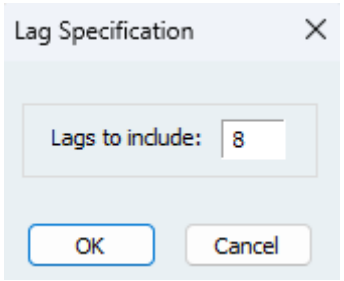
They disagree routinely. When they do, estimate at the competing orders and let the residual diagnostics decide – a lag length that leaves autocorrelation is not sufficient whatever the criterion says.

Sample-size reality check:

- $p = 4$ when working with quarterly data;
- $p = 12$ when working with monthly data;
- Practitioners keep np well under $T/3$.

Assessing lag length in EViews





No hard rules:

- set 12 or 24 lags for monthly data
- set 4 or 8 lags for quarterly data
- set 1 or 2 lags for annual data

Most criteria select one lag here

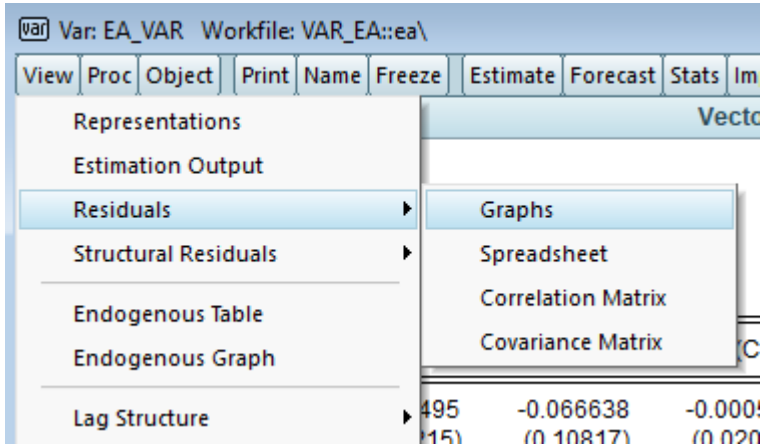
VAR Lag Order Selection Criteria
Endogenous variables: GDP_GAP REER_GAP DLOG(CPI)*100 RS_SHADOW
Exogenous variables: C
Date: 07/30/26 Time: 12:49
Sample: 2002Q1 2024Q4
Included observations: 91

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-722.8605	NA	101.8448	15.97496	16.08532	16.01948
1	-437.1212	540.0786	0.271323*	10.04662*	10.59846*	10.26925*
2	-424.0915	23.48198	0.290250	10.11190	11.10521	10.51264
3	-406.5303	30.10500	0.281944	10.07759	11.51237	10.65643
4	-394.6259	19.36103	0.311565	10.16760	12.04385	10.92455
5	-384.3096	15.87122	0.358750	10.29252	12.61023	11.22757
6	-373.4247	15.78910	0.411218	10.40494	13.16412	11.51810
7	-348.4269	34.06289*	0.349118	10.20718	13.40784	11.49845
8	-329.6908	23.88341	0.344314	10.14705	13.78918	11.61642

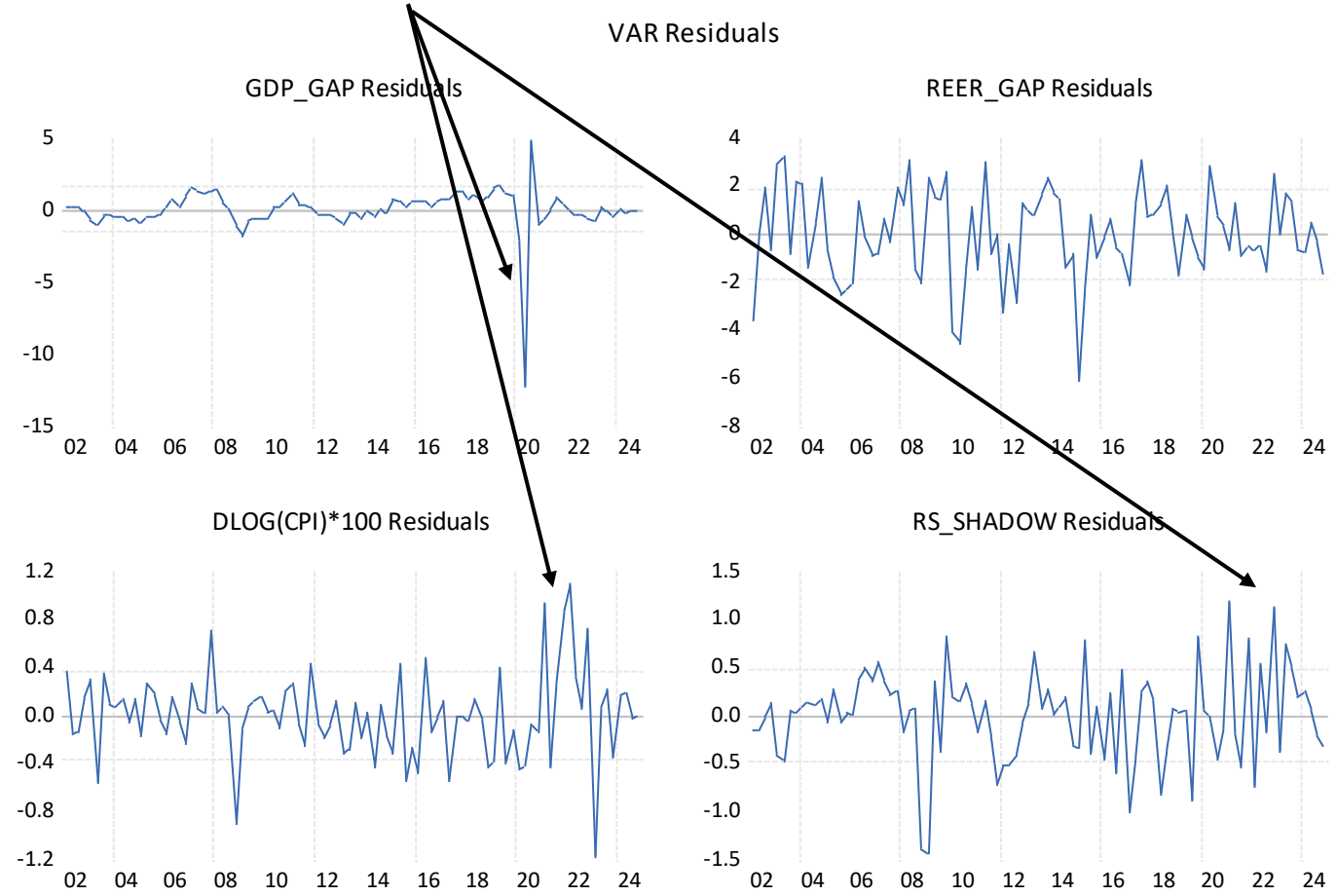
* indicates lag order selected by the criterion
LR: sequential modified LR test statistic (each test at 5% level)
FPE: Final prediction error
AIC: Akaike information criterion
SC: Schwarz information criterion
HQ: Hannan-Quinn information criterion

3. Residual diagnostics

VAR residuals:
Check whether they seem to be white noises



Residuals aren't white noises

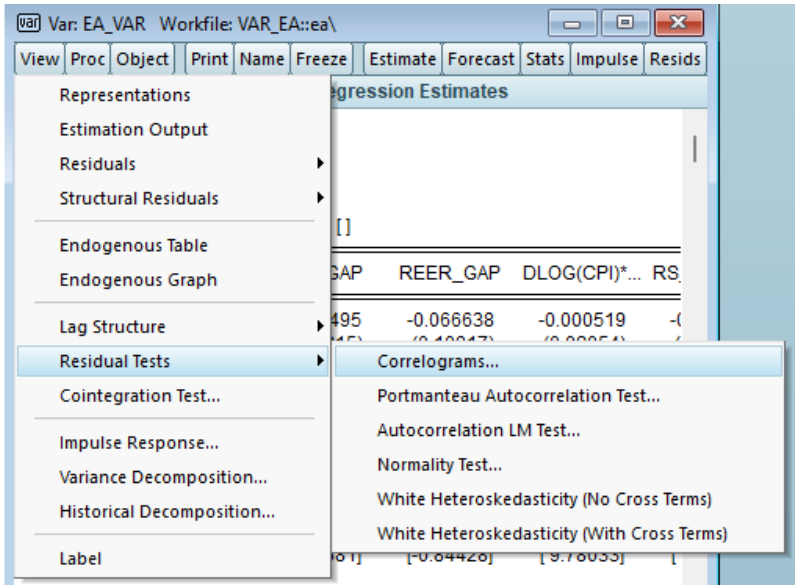


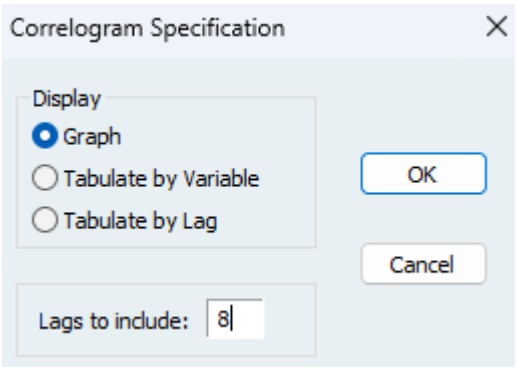
Residuals are correlated among themselves as expected for reduced-form VAR

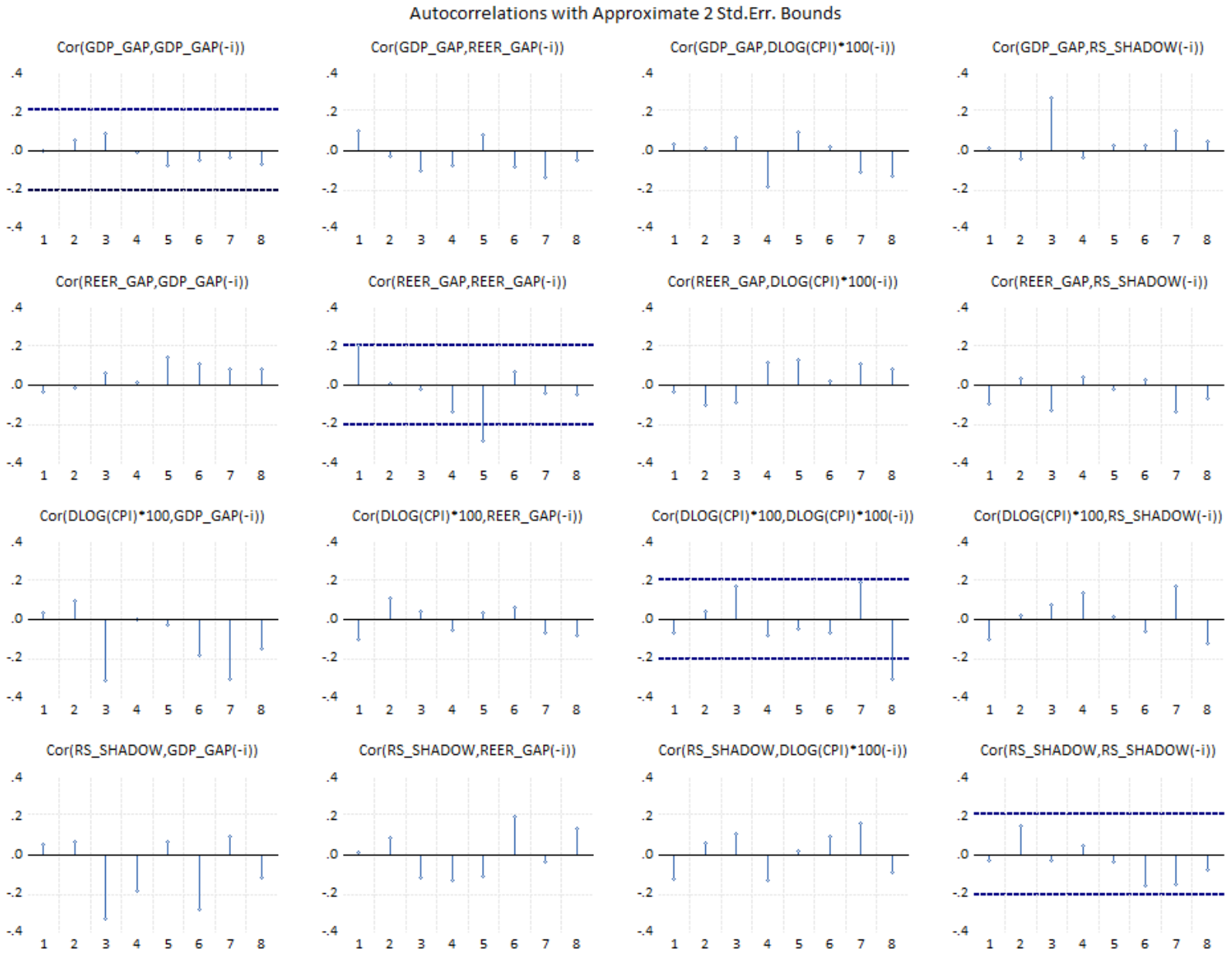
VAR Residual Correlation Matrix

	GDP_GAP	REER_GAP	DLOG(CPI)*100	RS_SHADOW
GDP_GAP	1.000000	-0.146798	0.164983	0.031721
REER_GAP	-0.146798	1.000000	0.012912	0.094467
DLOG(CPI)*100	0.164983	0.012912	1.000000	0.366544
RS_SHADOW	0.031721	0.094467	0.366544	1.000000

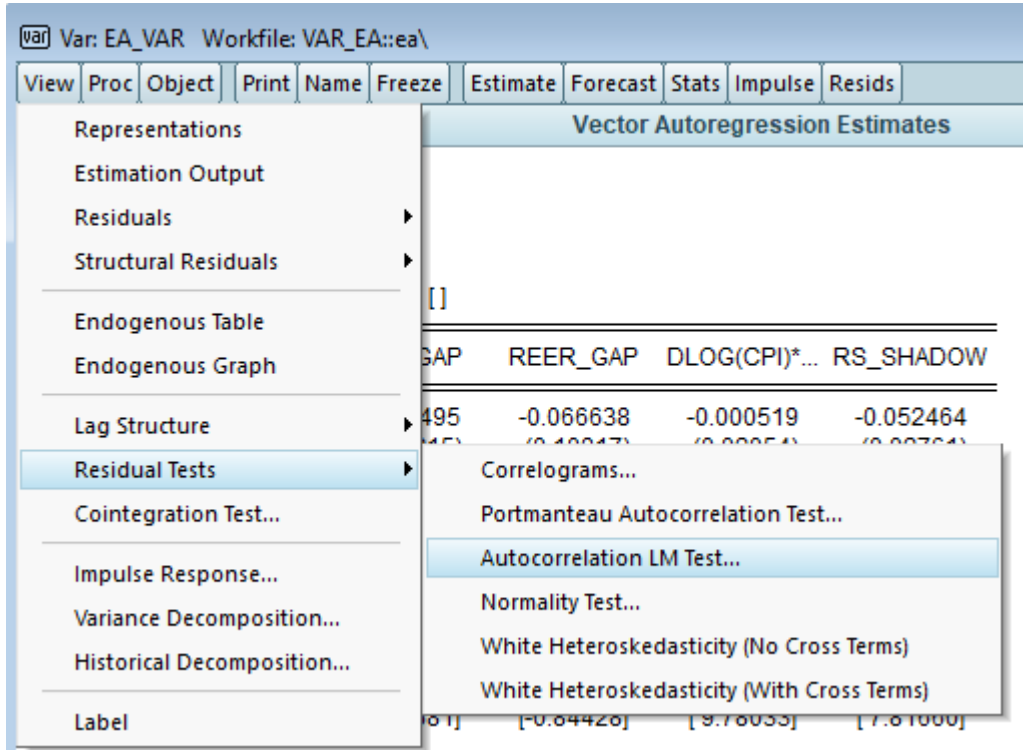
VAR residuals: Correlograms

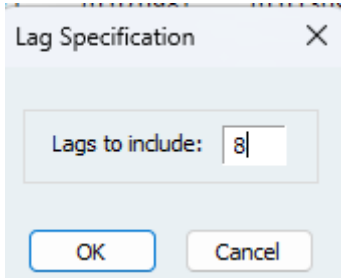






VAR residual tests. Autocorrelation LM test





H_0: *no serial correlation up to a chosen lag order*

Consequences of failing:

- dynamics are misspecified
- OLS is no longer consistent
- the impulse-responses and variance decompositions are invalid – the point estimates, not only their confidence bands

VAR residual tests.
Autocorrelation LM test results

VAR Residual Serial Correlation LM Tests
Date: 07/30/26 Time: 12:52
Sample: 2002Q1 2024Q4
Included observations: 92

Null hypothesis: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	20.12482	16	0.2147	1.273015	(16, 245.0)	0.2150
2	7.606161	16	0.9597	0.469227	(16, 245.0)	0.9598
3	38.98911	16	0.0011	2.562212	(16, 245.0)	0.0011
4	17.99651	16	0.3241	1.133530	(16, 245.0)	0.3245
5	14.39642	16	0.5692	0.900253	(16, 245.0)	0.5695
6	21.39287	16	0.1639	1.356682	(16, 245.0)	0.1642
7	40.95009	16	0.0006	2.701847	(16, 245.0)	0.0006
8	20.07954	16	0.2167	1.270035	(16, 245.0)	0.2170

Null hypothesis: No serial correlation at lags 1 to h

Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	20.12482	16	0.2147	1.273015	(16, 245.0)	0.2150
2	47.82502	32	0.0357	1.539711	(32, 281.9)	0.0361
3	69.70455	48	0.0220	1.508966	(48, 279.4)	0.0227
4	82.73270	64	0.0576	1.334778	(64, 268.5)	0.0608
5	92.59436	80	0.1587	1.179975	(80, 254.9)	0.1695
6	118.3320	96	0.0607	1.275721	(96, 240.2)	0.0705
7	159.1353	112	0.0023	1.537504	(112, 225.0)	0.0035
8	183.3608	128	0.0010	1.571052	(128, 209.5)	0.0019

*Edgeworth expansion corrected likelihood ratio statistic.

Overall conclusion :

The VAR **fails the no-serial-correlation null** – the residuals are not white noise.

Possible solutions:

- COVID-19 shock – add impulse dummies for 2020.
- Consider adding more lags (2 – 4): cleaning 2020 can change which lag length the criteria prefer.
- Think about exogenous regressors (oil prices, GSCPI)

VAR residuals: Homoskedasticity and normality

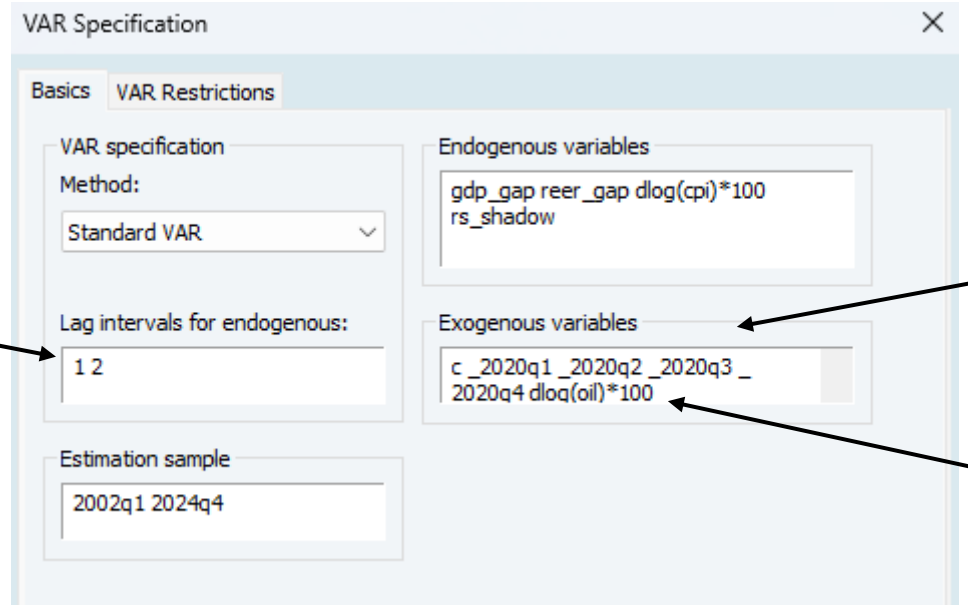
White heteroskedasticity test – tests the null of homoskedastic, constant-variance residuals *(the no-cross-terms version is the practical default in VARs with many regressors)*.

Failing it does **not** break consistency, but standard errors are wrong. For macro data with crisis episodes this test frequently fails, and that's usually acceptable as long as you use robust inference for IRFs (bootstrapped SE).

Normality test – Jarque–Bera type, testing whether residuals have Normal distribution. Normality is not needed for consistency, and by asymptotic theory your estimators and tests are still valid in large samples without it. In practice macro residuals are almost always non-normal.

VAR residuals: Fixing the problems (1)

Add more lags to mitigate the autocorrelation issue



Add impulse dummies for the pandemic period

Add oil prices as important factor of inflation

VAR residuals: Fixing the problems (2)

The residuals look much more like white noise

The autocorrelation problem has been eliminated

VAR Residual Serial Correlation LM Tests
Date: 07/30/26 Time: 13:43
Sample: 2002Q1 2024Q4
Included observations: 92

Null hypothesis: No serial correlation at lag h

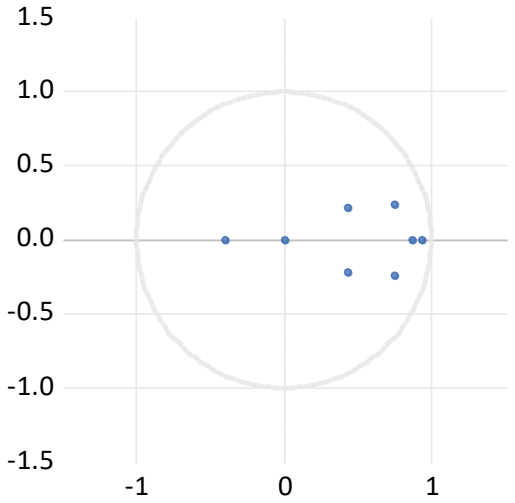
Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	17.38996	16	0.3608	1.094838	(16, 217.5)	0.3613
2	24.80752	16	0.0733	1.588254	(16, 217.5)	0.0735
3	13.21664	16	0.6569	0.824310	(16, 217.5)	0.6572
4	13.46600	16	0.6384	0.840333	(16, 217.5)	0.6388
5	17.27751	16	0.3679	1.087483	(16, 217.5)	0.3683
6	14.35214	16	0.5725	0.897420	(16, 217.5)	0.5729
7	27.36145	16	0.0376	1.761949	(16, 217.5)	0.0378
8	13.52424	16	0.6341	0.844079	(16, 217.5)	0.6345

Null hypothesis: No serial correlation at lags 1 to h

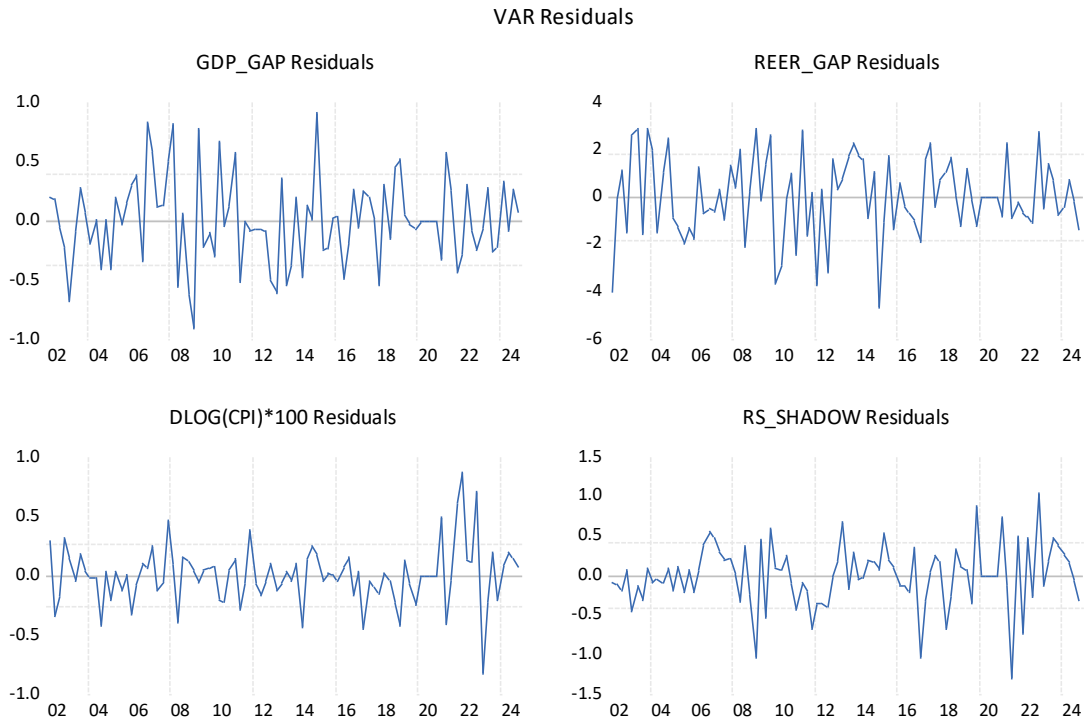
Lag	LRE* stat	df	Prob.	Rao F-stat	df	Prob.
1	17.38996	16	0.3608	1.094838	(16, 217.5)	0.3613
2	43.16125	32	0.0900	1.382092	(32, 248.7)	0.0910
3	55.09143	48	0.2242	1.164882	(48, 244.7)	0.2285
4	70.08169	64	0.2810	1.106860	(64, 233.3)	0.2910
5	84.68477	80	0.3388	1.063571	(80, 219.4)	0.3582
6	101.3224	96	0.3354	1.056833	(96, 204.6)	0.3679
7	134.7750	112	0.0704	1.244252	(112, 189.3)	0.0935
8	150.0241	128	0.0892	1.200685	(128, 173.7)	0.1311

*Edgeworth expansion corrected likelihood ratio statistic.

Inverse Roots of AR Characteristic Polynomial

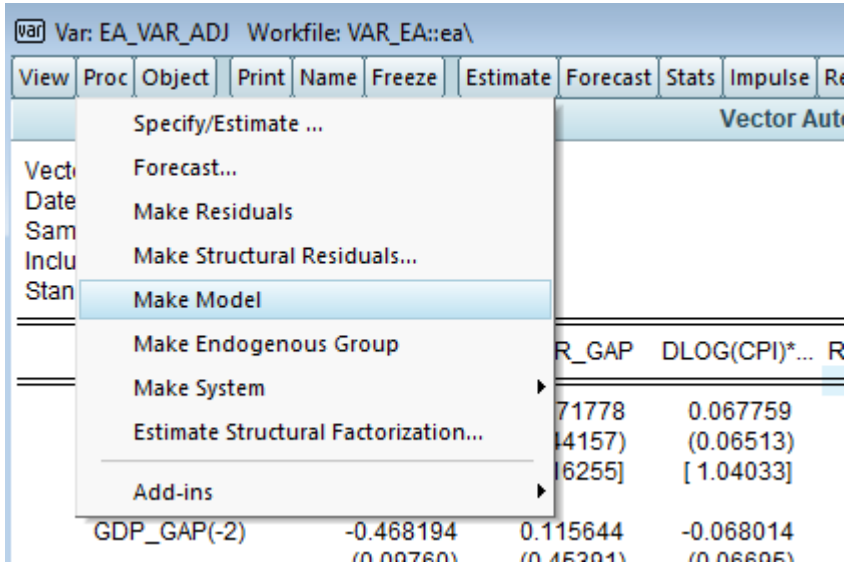


The VAR remains stationary



4. Forecasting with VARs

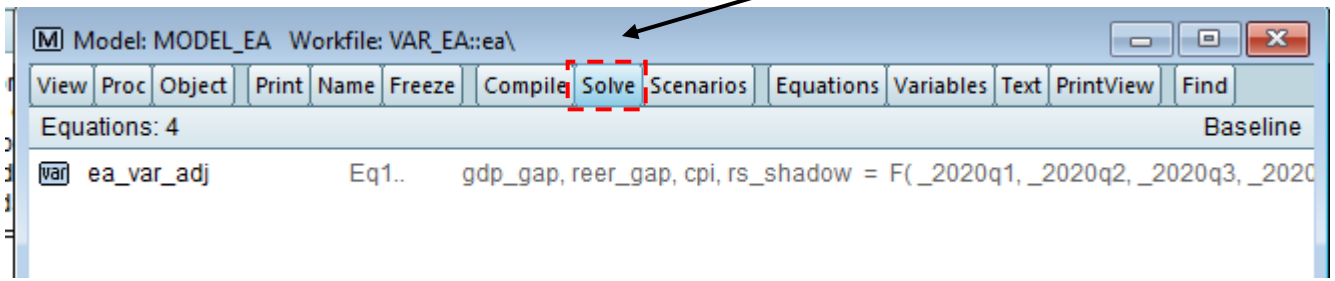
Creating a model from a VAR

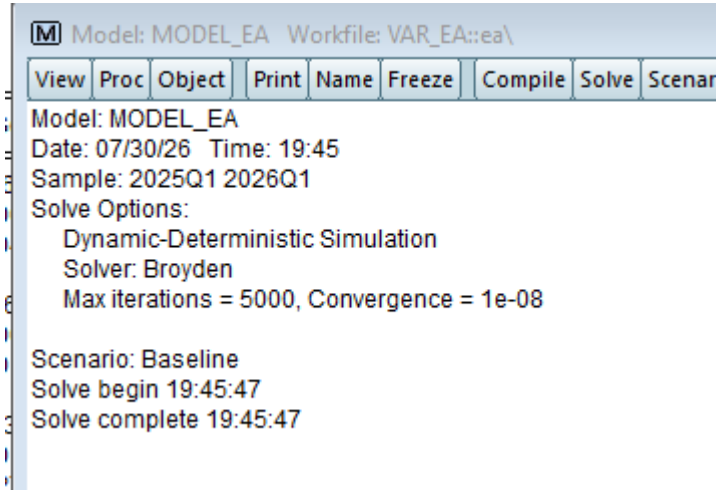




Solving the model

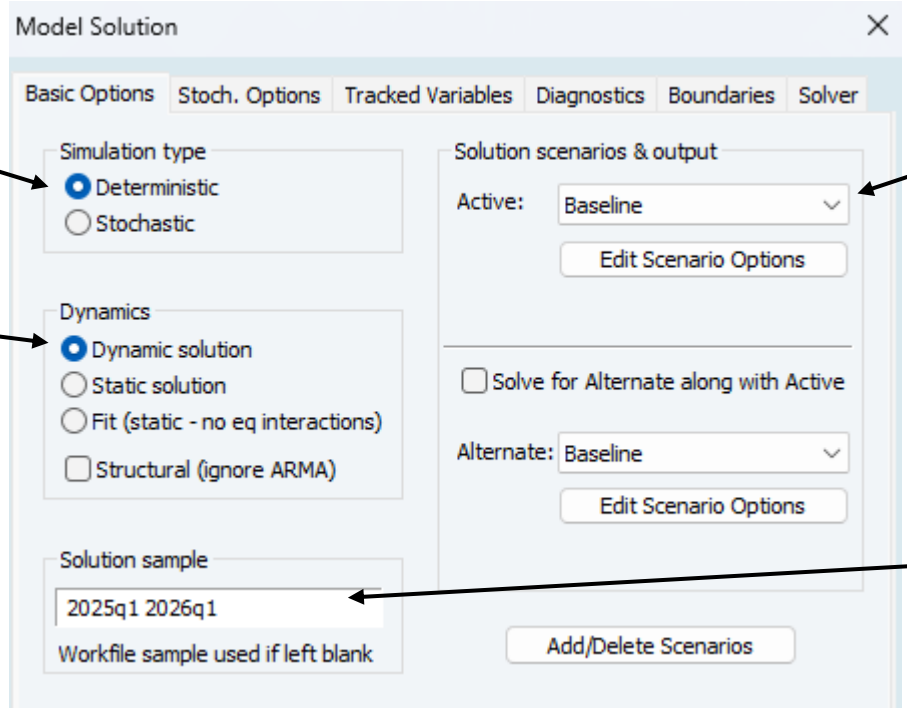
Click on "Solve"





Only the mean forecast *(no uncertainty included)*

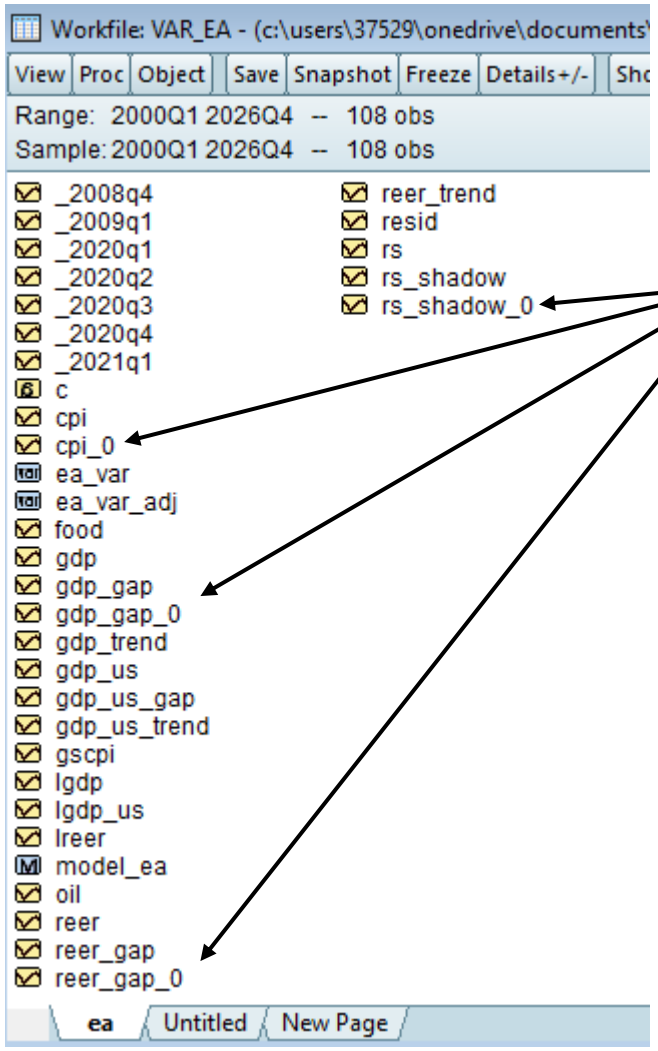
Iterative forecast after the first prediction period *(endogenous variables values are not know after period t; exogenous variables values must be prepared and inserted into the series)*



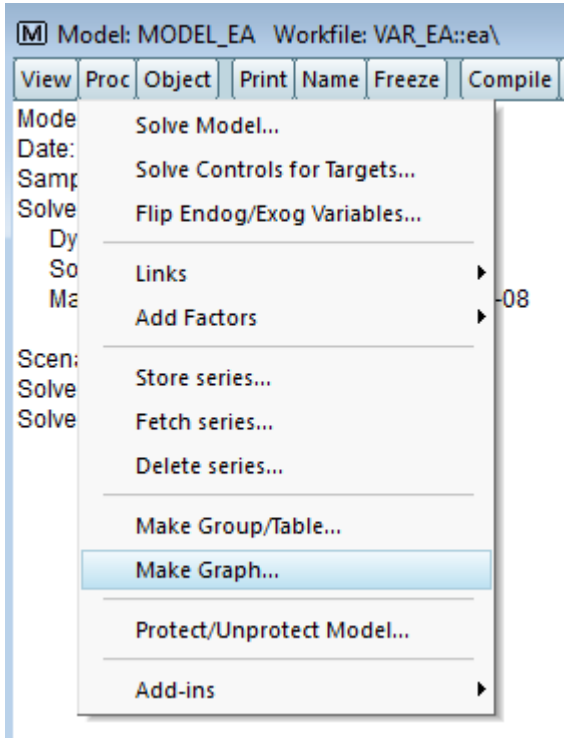
Choose the scenario *(Baseline is a default option)*

Specify the forecasting period

Forecasting results

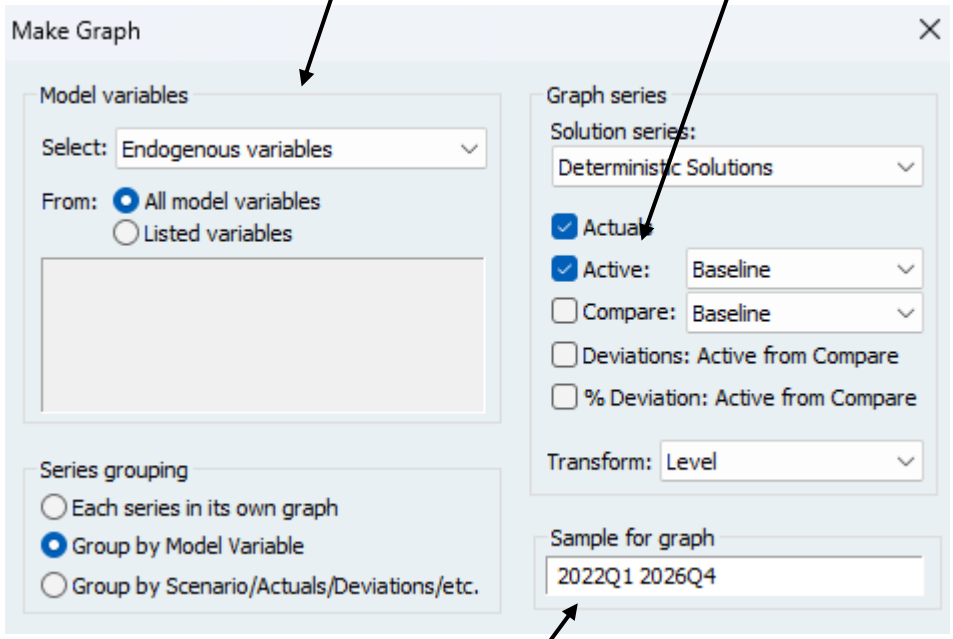


EViews saves the forecast for all of the endogenous variables in the workfile with the _0 extension.



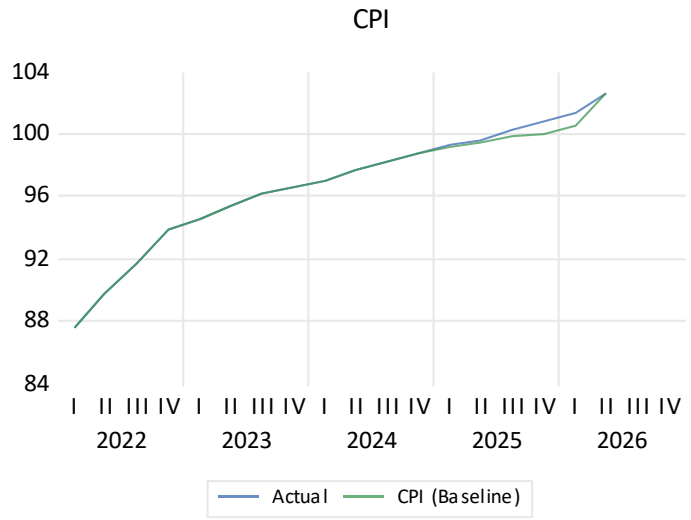
Select the variables to show

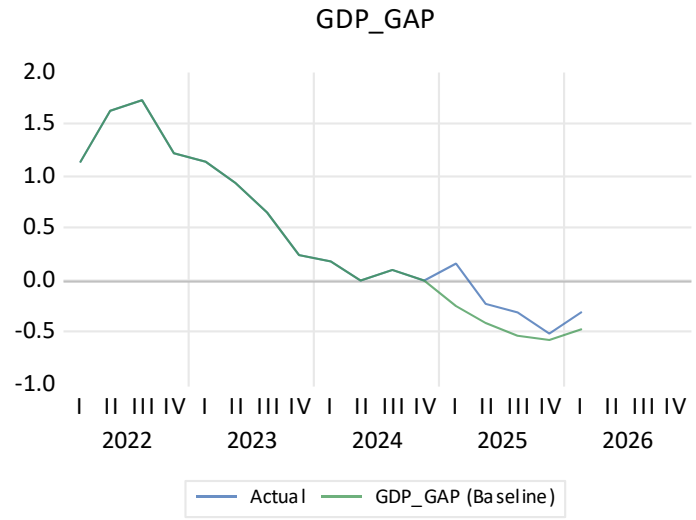
Choose scenarios to compare

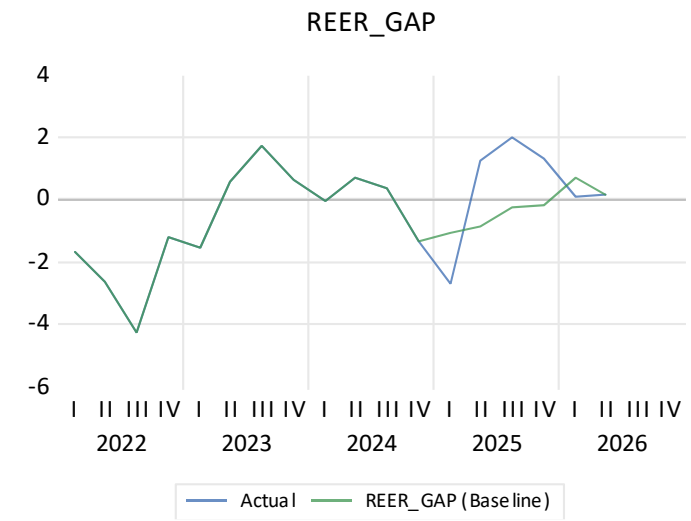


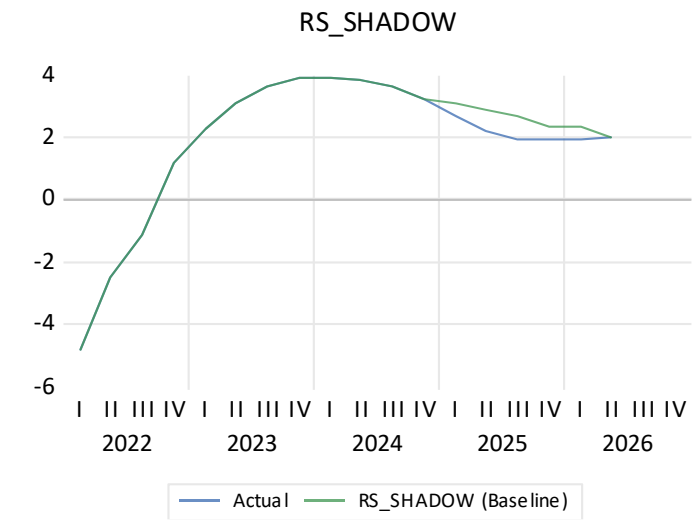
Specify the sample for the graph

Deterministic forecast path









Stochastic forecast

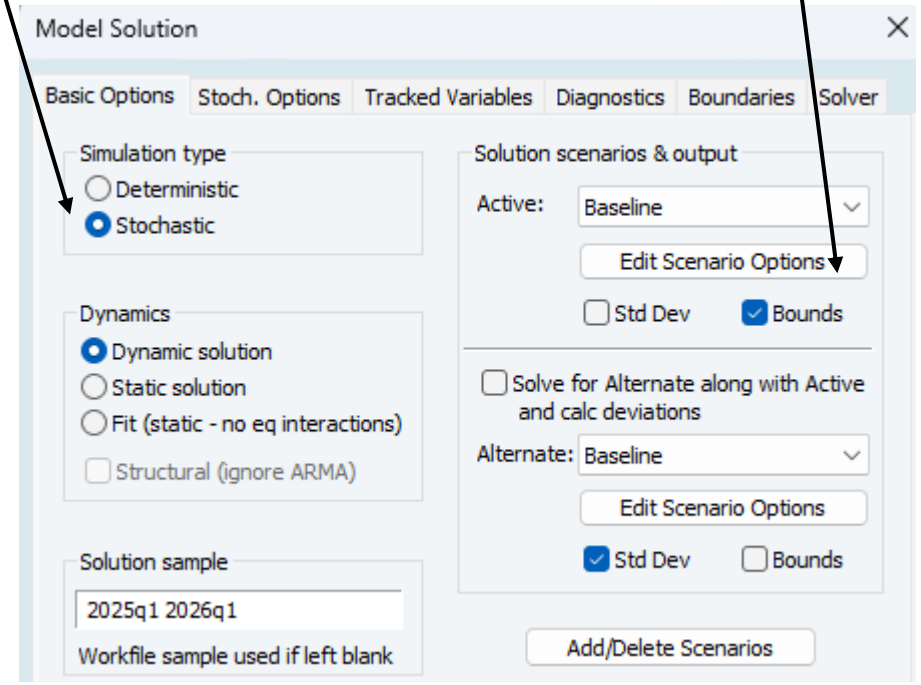
Deterministic forecast assumes that our stochastic equations would hold exactly over the forecast period.

In reality, these relationships do not hold exactly because of:

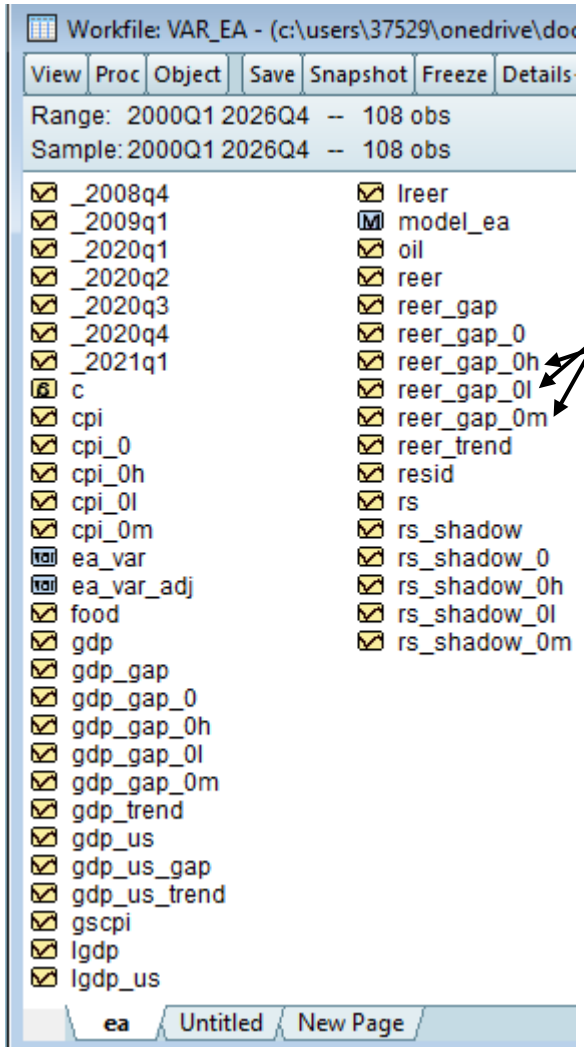
- random disturbances;
- our coefficients are estimated rather than being some known or predetermined values.

Allow us to add uncertainty by adding error bounds to our projections

By default – 95% confidence interval

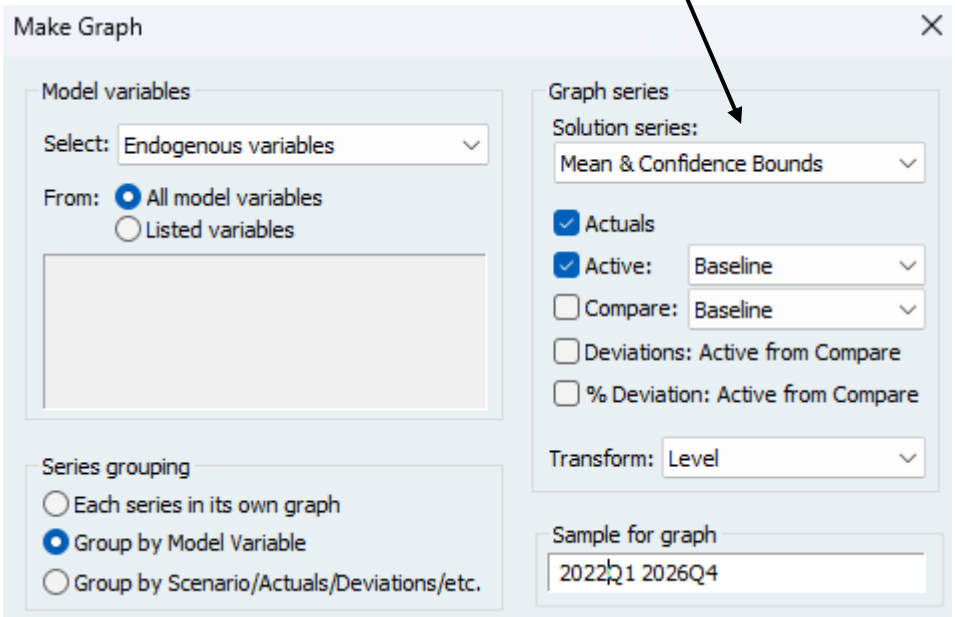


Stochastic forecast results

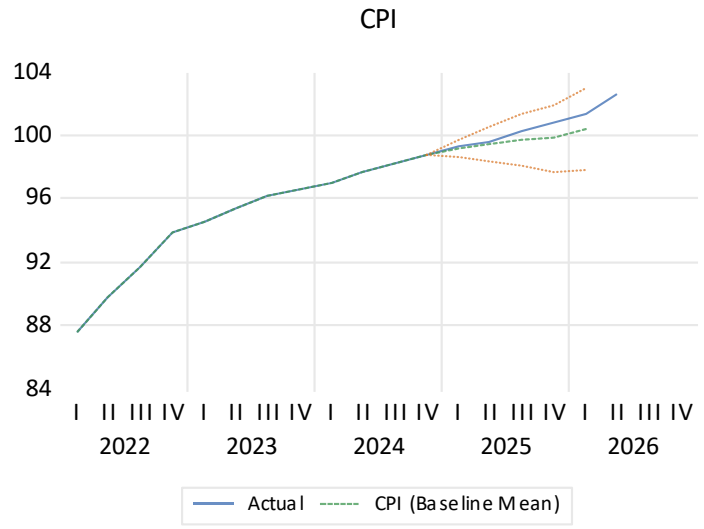


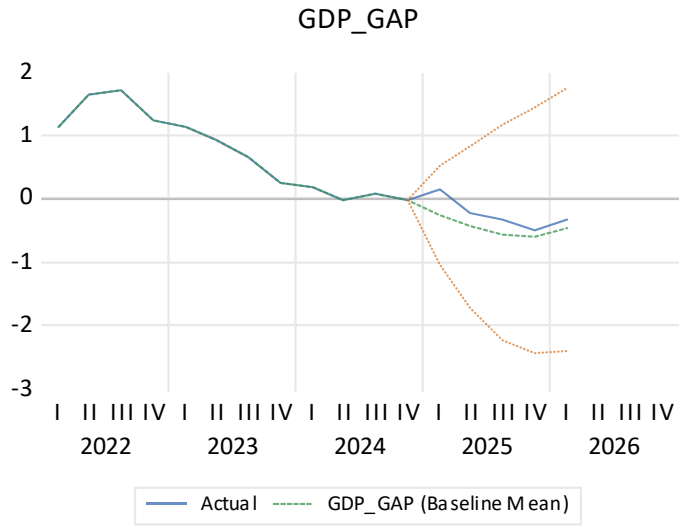
EViews has added additional series:
_0m (mean forecast),
_0l (lower bound),
_0h (upper bound)

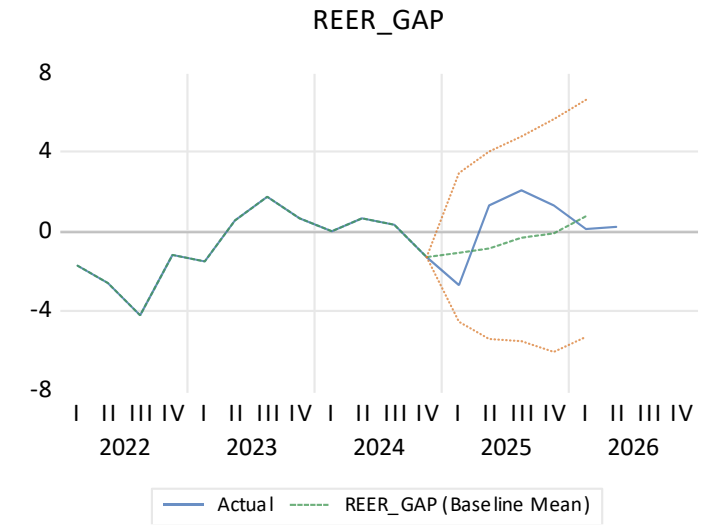
We can view the forecast with the confidence bands

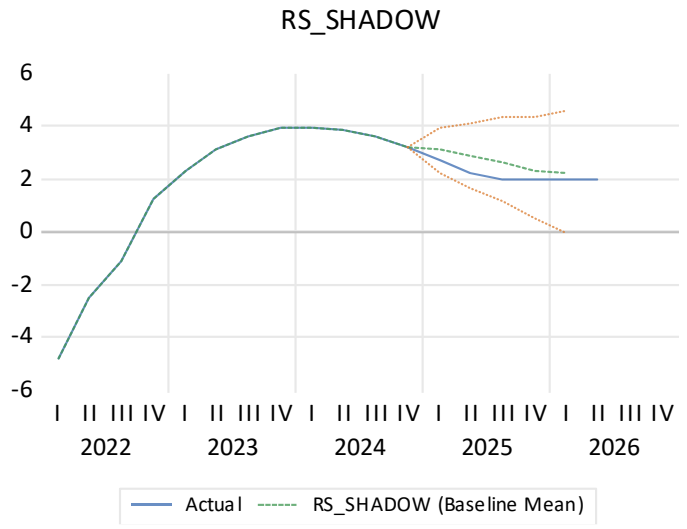


Stochastic forecast path



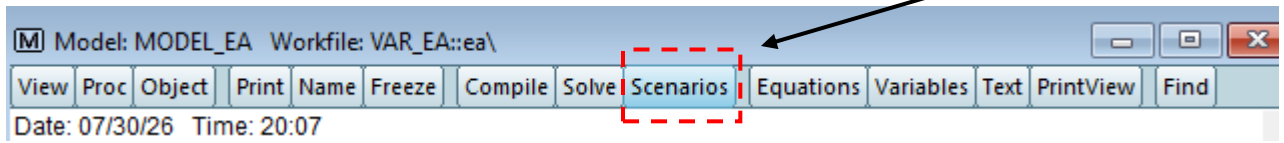




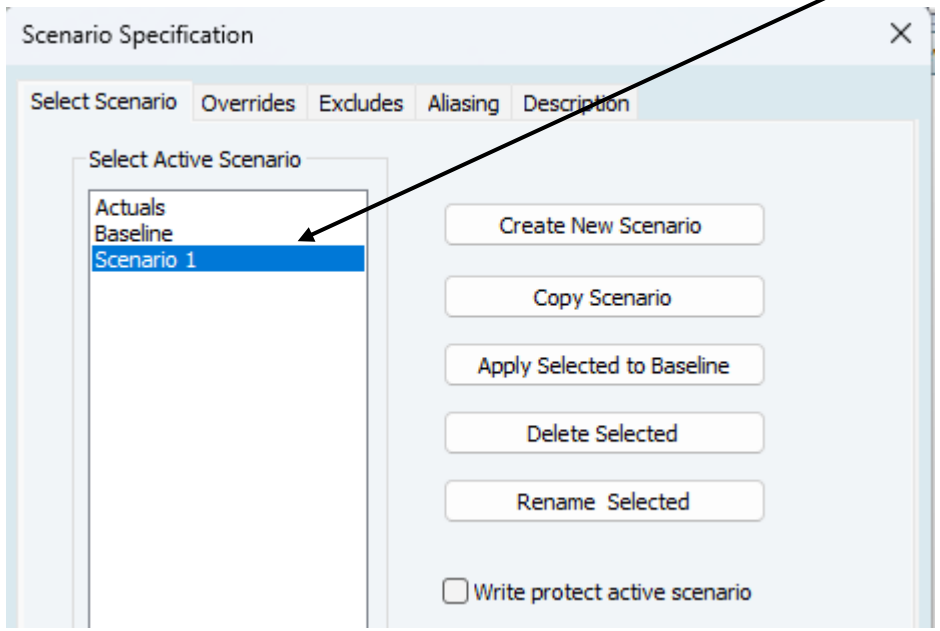


Creating alternative scenarios

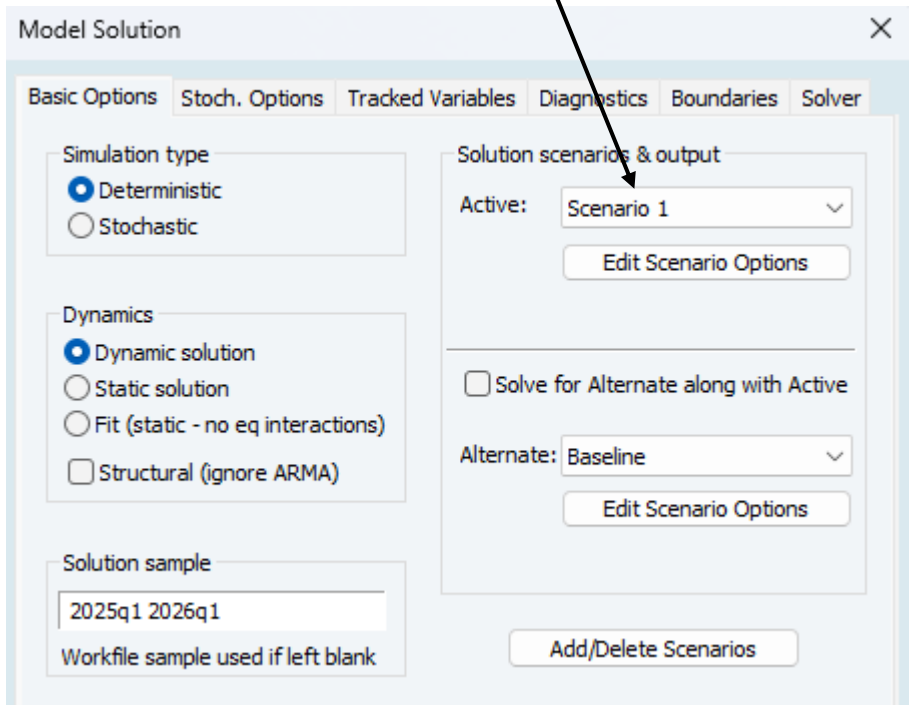
Click on “Scenarios"



Select “Scenario 1"

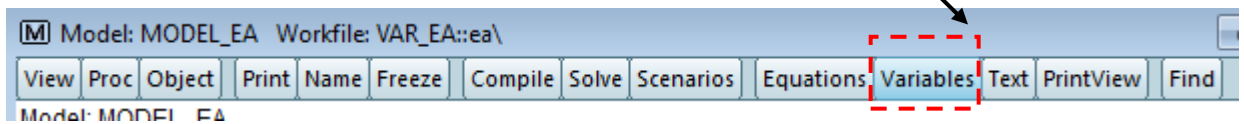


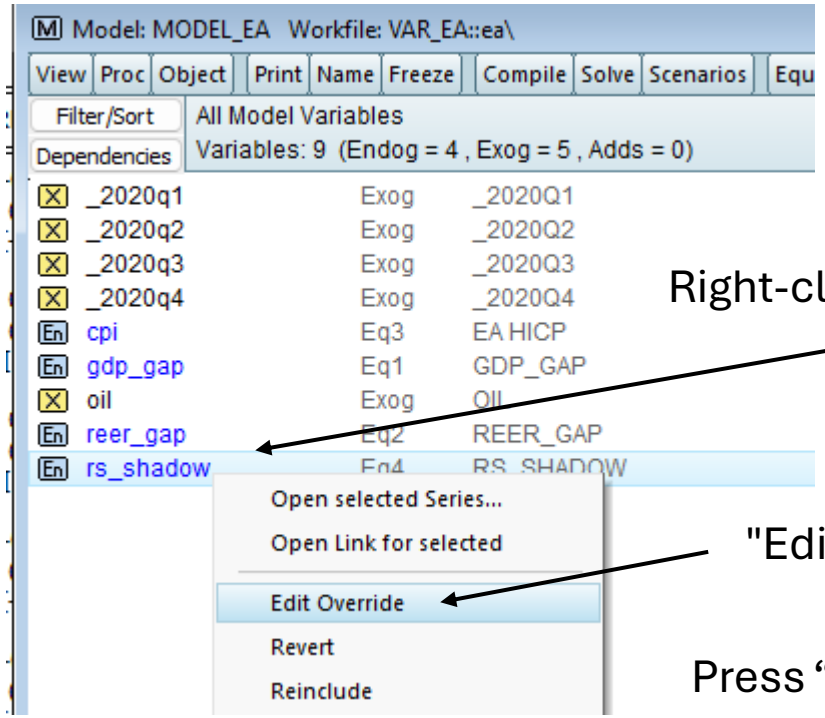
Solve the model for Scenario 1



Specifying the alternative scenario.
More aggressive interest rate reduction path

Click on “Variables"

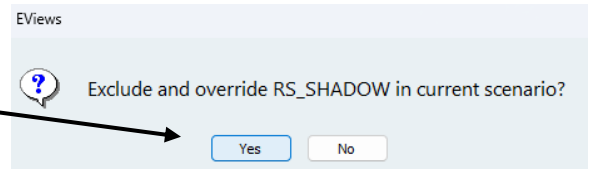




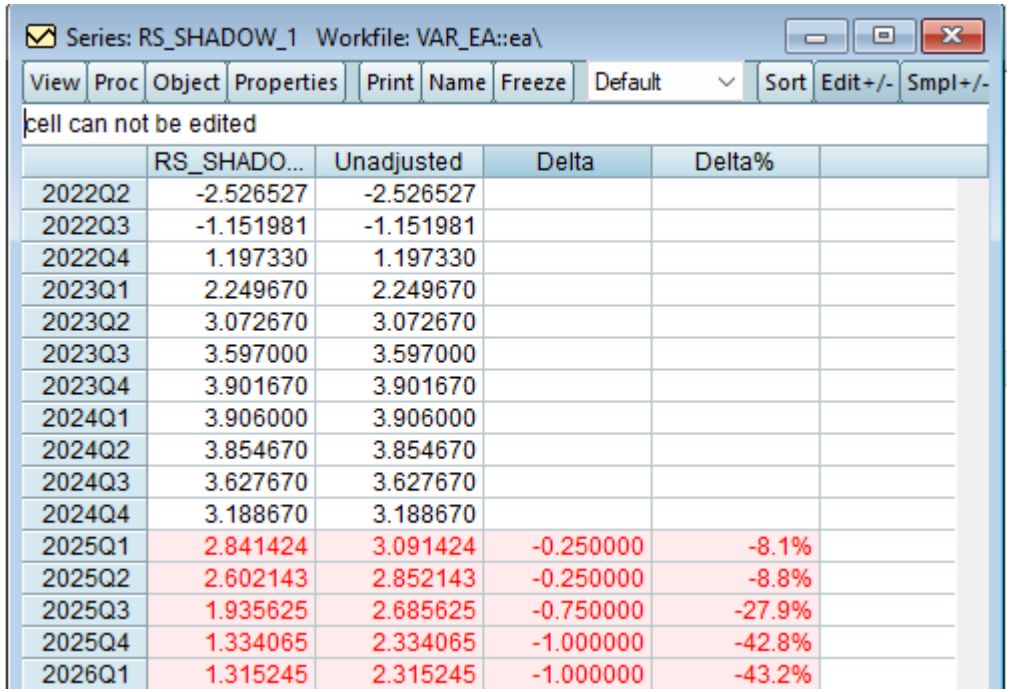
Right-click the variable

"Edit, override"

Press “Yes”



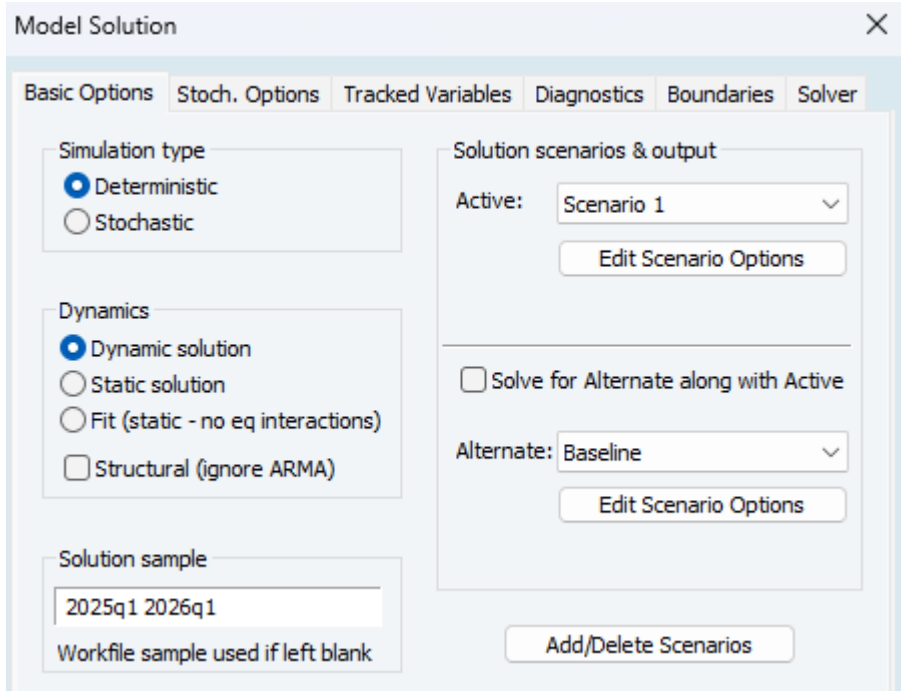
Specify the delta, which is like a change, or a delta percent, which would be a percentage change, compared to unadjusted forecast



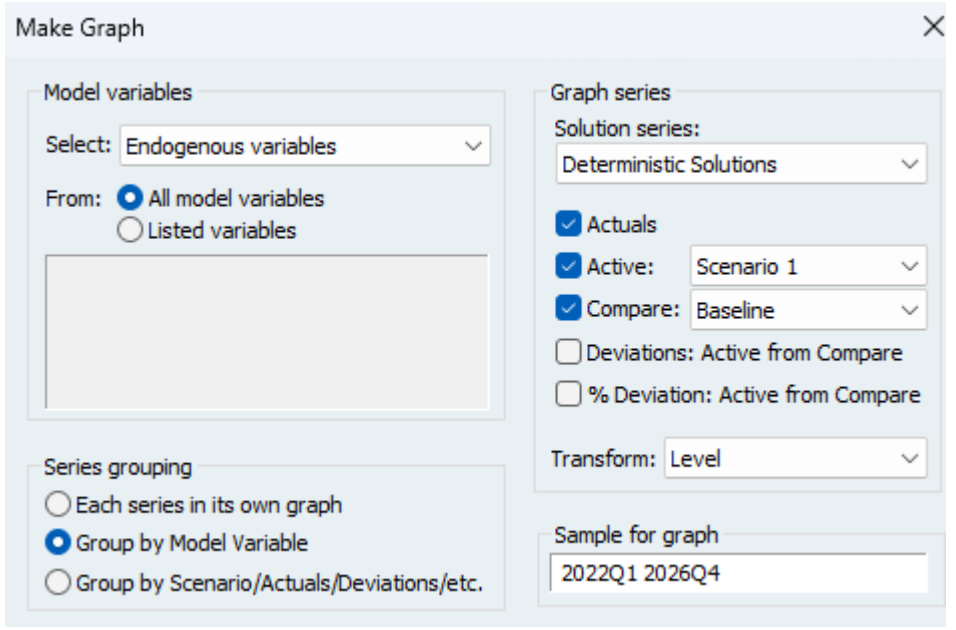
	RS_SHADO...	Unadjusted	Delta	Delta%
2022Q2	-2.526527	-2.526527		
2022Q3	-1.151981	-1.151981		
2022Q4	1.197330	1.197330		
2023Q1	2.249670	2.249670		
2023Q2	3.072670	3.072670		
2023Q3	3.597000	3.597000		
2023Q4	3.901670	3.901670		
2024Q1	3.906000	3.906000		
2024Q2	3.854670	3.854670		
2024Q3	3.627670	3.627670		
2024Q4	3.188670	3.188670		
2025Q1	2.841424	3.091424	-0.250000	-8.1%
2025Q2	2.602143	2.852143	-0.250000	-8.8%
2025Q3	1.935625	2.685625	-0.750000	-27.9%
2025Q4	1.334065	2.334065	-1.000000	-42.8%
2026Q1	1.315245	2.315245	-1.000000	-43.2%

Alternative scenario results (1)

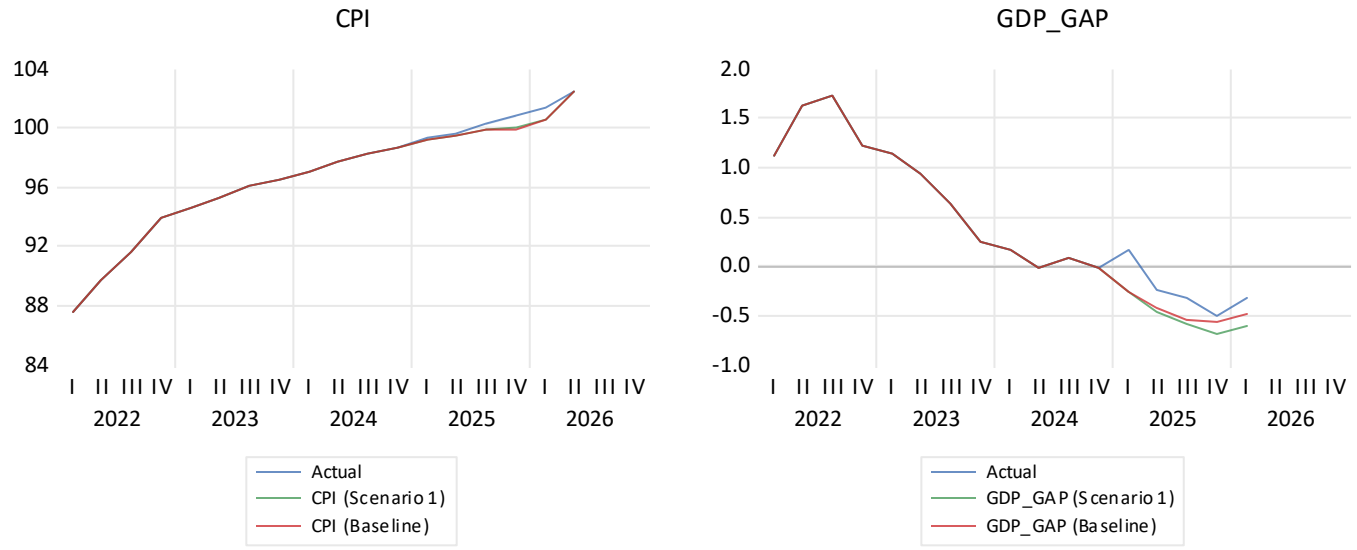
Solve for the alternative scenario

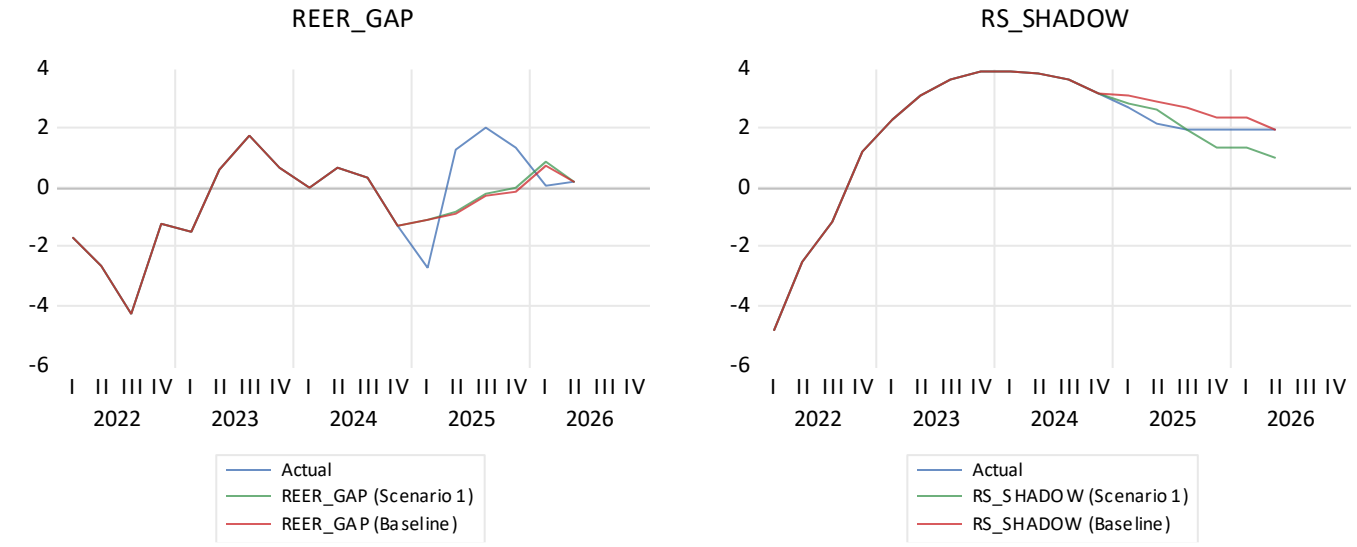


Compare with actuals and the baseline



Alternative scenario results (2)





5. SVARs

Why the reduced form cannot answer the policy question

The problem:

1. The central bank sees inflation coming …
2. … and raises the policy rate in advance.
3. Inflation then rises anyway, for the reasons the bank foresaw.

Wrong conclusion: tightening is inflationary.
Reality: endogenous reaction to expected inflation.

The reduced-form residual e_{rt} is a forecast error, not a policy shock. It bundles the part of the rate move that responded to conditions the bank observed together with the part that was genuinely unanticipated.

What identification is. Recovering economically interpretable shocks from the forecast errors by imposing restrictions the data cannot supply.

Writing down the structural VAR (SVAR)

A SVAR(1) with independent structural shocks $\mathrm{u_t}$:

$$B\mathrm{y_t} = \Gamma_0 + \Gamma_1 \mathrm{y_{t-1}} + \mathrm{u_t}$$

Two variables, inflation (π) and interest rate (r), each allowed to respond to the other **within** the period as well as with a lag.

$$\pi_t = \gamma_{10} - b_{12} r_t + \gamma_{11}\pi_{t-1} + \gamma_{12} r_{t-1} + u_{\pi t},$$

$$r_t = \gamma_{20} - b_{21}\pi_t + \gamma_{21}\pi_{t-1} + \gamma_{22} r_{t-1} + u_{rt}$$

In matrix form:

$$\underbrace{\begin{bmatrix} 1 & b_{12} \\ b_{21} & 1 \end{bmatrix}}_{B} \begin{bmatrix} \pi_t \\ r_t \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \gamma_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} \pi_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} u_{\pi t} \\ u_{rt} \end{bmatrix}$$

What OLS actually estimates

Pre-multiply by B^{-1} and the simultaneity disappears:

$$B^{-1}By_t = B^{-1}\Gamma_0 + B^{-1}\Gamma_1 y_{t-1} + B^{-1}u_t$$

$$y_t = a_0 + A_1 y_{t-1} + e_t$$

$$e_t = B^{-1}u_t$$

This is estimable – every right-hand-side variable is predetermined.

The price: each reduced-form error is a blend of both structural shocks.

$$e_{\pi t} = \frac{u_{\pi t} - b_{12}u_{rt}}{1 - b_{12}b_{21}}$$

$$e_{rt} = \frac{u_{rt} - b_{21}\,u_{\pi t}}{1 - b_{12}b_{21}}$$

The counting problem

The reduced-form VAR:

$$\begin{bmatrix}\pi_t \\ r_t\end{bmatrix} = \begin{bmatrix}a_{10} \\ a_{20}\end{bmatrix} + \begin{bmatrix}a_{11} & a_{12} \\ a_{21} & a_{22}\end{bmatrix}\begin{bmatrix}\pi_{t-1} \\ r_{t-1}\end{bmatrix} + \begin{bmatrix}e_{\pi t} \\ e_{rt}\end{bmatrix}$$

We have 9 parameters: 6 coefficients, 2 variances and 1 covariance of the residuals

What the structural model asks for: 10 parameters – 8 coefficients and 2 variances:

$$\begin{bmatrix}1 & b_{12} \\ b_{21} & 1\end{bmatrix}\begin{bmatrix}\pi_t \\ r_t\end{bmatrix} = \begin{bmatrix}\gamma_{10} \\ \gamma_{20}\end{bmatrix} + \begin{bmatrix}\gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22}\end{bmatrix}\begin{bmatrix}\pi_{t-1} \\ r_{t-1}\end{bmatrix} + \begin{bmatrix}u_{\pi t} \\ u_{rt}\end{bmatrix}$$

One **restriction** has to come from outside the data.

Where the restriction goes

Restrictions are normally placed on *B*, because that is where the economics is: each off-diagonal entry is a claim about what moves what within the period.

$$\begin{bmatrix} 1 & \mathbf{0} \\ b_{21} & 1 \end{bmatrix} \begin{bmatrix} \pi_t \\ r_t \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \gamma_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} \pi_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} u_{\pi t} \\ u_{rt} \end{bmatrix}$$

Set $b_{12} = 0$, and you have asserted that inflation does not respond to the policy rate within the quarter – policy works with a lag – while the rate still responds to inflation immediately.

$$\pi_t = \gamma_{10} + \gamma_{11}\pi_{t-1} + \gamma_{12}r_{t-1} + u_{\pi t},$$

$$r_t = \gamma_{20} - b_{21}\pi_t + \gamma_{21}\pi_{t-1} + \gamma_{22}r_{t-1} + u_{rt}$$

Count again: nine knowns, nine remaining unknowns. Exactly identified.

Recursive ordering

A recursive ordering is a set of *n(n−1)/2* zeros placed to make *B* triangular.

Economically: a sequential chain, where each variable may respond immediately to those before it and only with a lag to those after.

$$\begin{bmatrix} e_{\pi t} \\ e_{rt} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \mathbf{0} \\ -b_{21} & 1 \end{bmatrix}}_{B^{-1}} \begin{bmatrix} u_{\pi t} \\ u_{rt} \end{bmatrix}$$

The ordering is a decision, not a discovery. Put the slowest-moving variable first.

A three-variable recursive ordering with *y* (output), *π* and *r*:

$$\begin{bmatrix} e_{yt} \\ e_{\pi t} \\ e_{rt} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0} & \mathbf{0} \\ -b_{21} & 1 & \mathbf{0} \\ -b_{31} + b_{21}b_{32} & -b_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{yt} \\ u_{\pi t} \\ u_{rt} \end{bmatrix}$$

Other issues

- If correlations between the errors are low, the order is irrelevant. But usually, correlations are strong and the order matters.
- Deciding the ordering (restrictions) to be imposed is crucial. Use economic intuition to decide it.

Other strategies:

- **Short-run restrictions, non-recursive.** Still zeros on *contemporaneous* relationships (matrix A), but not forced into a triangular order.
- **Long-run restrictions.** Restrict the *cumulative/permanent* effect rather than the impact effect.
- **Sign restrictions.** Instead of setting things to zero, you restrict the *sign* of impulse responses for a few periods
- **Data-driven approaches.**

Summary

1. **Goal.** Structural shocks that can be interpreted economically.

2. **Obstacle.** OLS gives forecast errors, which are blends of those shocks.

3. **Fix.** Restrict the contemporaneous matrix, using economics rather than statistics.

How many restrictions?
$n(n-1)/2$

4. **Result.** Structural shocks and parameters recovered from the reduced-form estimates.

SVAR identification is about imposing restrictions on matrix B.

6. Impulse responses & variance decomposition

Two questions

Identification gave us **independent structural shocks** u – a clean demand shock, supply shock, policy shock, and so on. Now we ask two things:

1 Impulse Response (IRF)

"What happens after a shock?"

The dynamic path of every variable following one shock – direction, size, and how long it lasts.

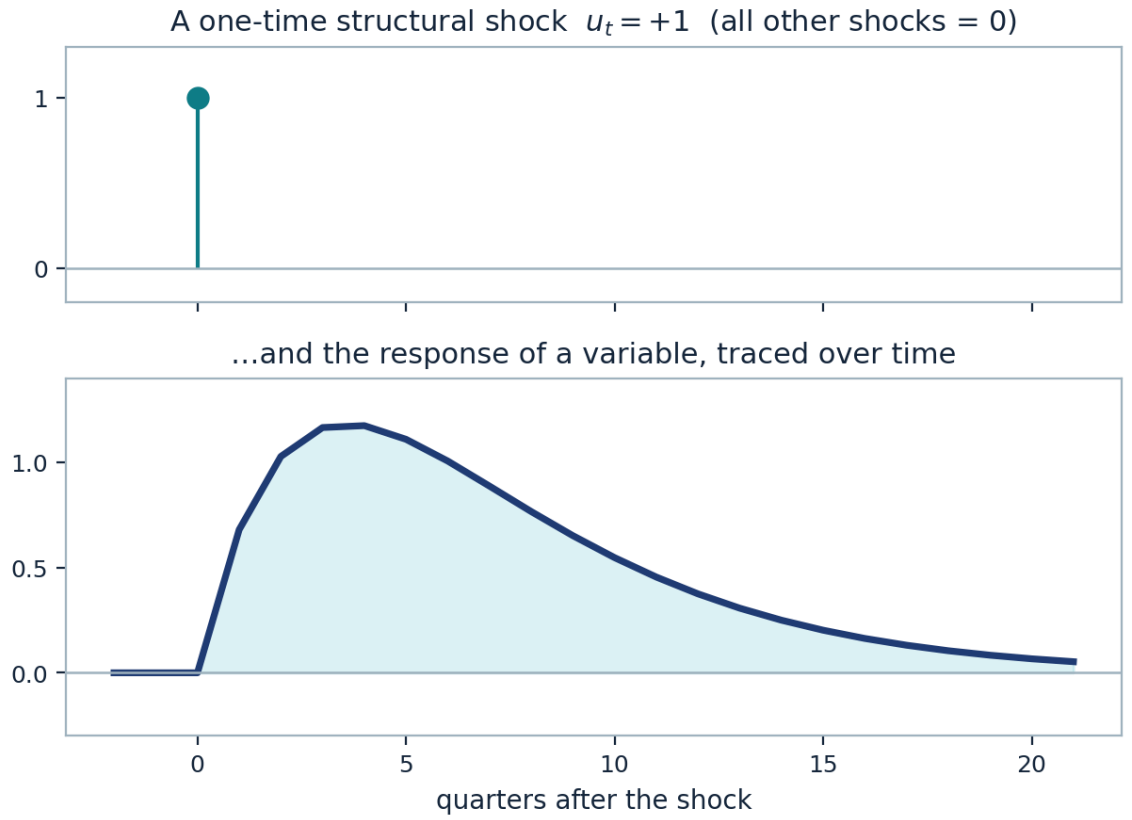
2 Variance Decomposition

"How much does each shock matter?"

The share of a variable's forecast uncertainty explained by each structural shock.

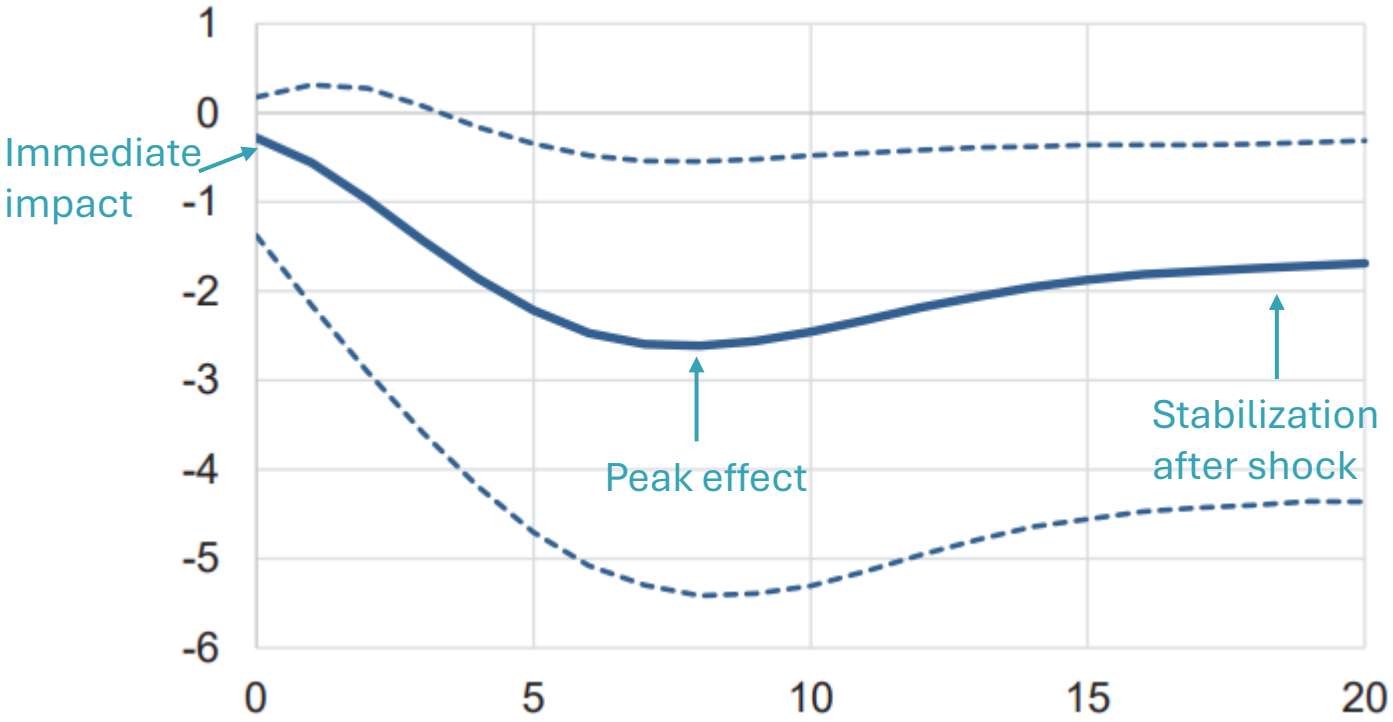
The impulse-response thought experiment

- Give the system a single, one-time “poke” – one structural shock of size +1.
- Hold all the other shocks at zero.
- Trace how every variable reacts – this quarter, next quarter, and on into the future.
- The reaction **spreads over time** because each variable depends on the lagged values of all the others.



Reading the impulse-response function (IRF)

Response of loans (%) to a monetary policy shock in Poland



- **Sign at impact** – Which way does the variable move right away?
- **Peak effect** – How big does the effect get, and when?
- **Persistence** – How many quarters until the impact of the shock dies out?
- **Bands** – The shaded region is the uncertainty around the estimate.

Notes: Monetary policy shock equals to a 1 p.p. rise in the interest rate. Horizontal axes – quarters after the shock. The dashed lines denote a 68% error band.
Source: Greszta, M. et al. (2023). Monetary policy transmission mechanism in Poland. What do we know in 2023? NBP Working Papers 365, Narodowy Bank Polski.

Where the IRFs come from?

- A stable (stationary) VAR can be rewritten as the present plus an infinite echo of past structural shocks:

$$\text{SVAR} \longrightarrow y_t = \mu + \sum_{i=0}^{\infty} \Theta_i \, u_{t-i} \longrightarrow \Theta_i = \Psi_i B^{-1} \longleftarrow y_t = \mu + \sum_{i=0}^{\infty} \Psi_i e_{t-i} \longleftarrow \text{Reduced-form VAR}$$

- The coefficient matrices Θ_0, Θ_1, Θ_2, … ARE the impulse responses: Θ_0 is the impact effect, Θ_1 the effect after one period, and so on.

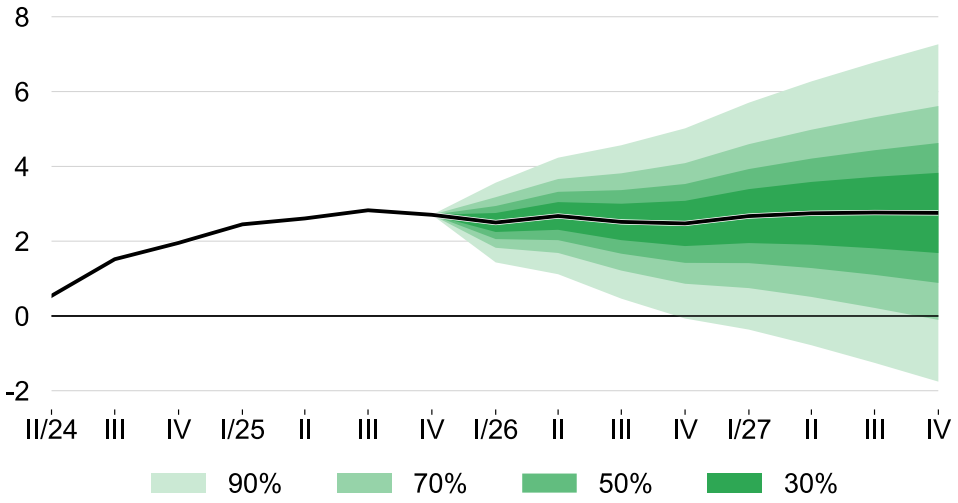
$$\text{IRF}_h = \frac{\partial y_{t+h}}{\partial u_t} = \Theta_h$$

Plotting Θ_0, Θ_1, Θ_2, … against the horizon draws the impulse response.

From “What Happens?” to “How Much?”

- The IRF tells you the **shape** of a shock's effect.
- But it doesn't say how **important** that shock is relative to the others.
- When we forecast ahead, we don't know the future shocks – so the forecast has error, and that error grows with the horizon.
- Variance decomposition splits that forecast uncertainty into **shares** by shock.

GDP forecast uncertainty: future shocks are unknown



Note: The confidence intervals of the GDP growth forecast reflect the predictive power of past forecasts. They are symmetric and linearly widening.

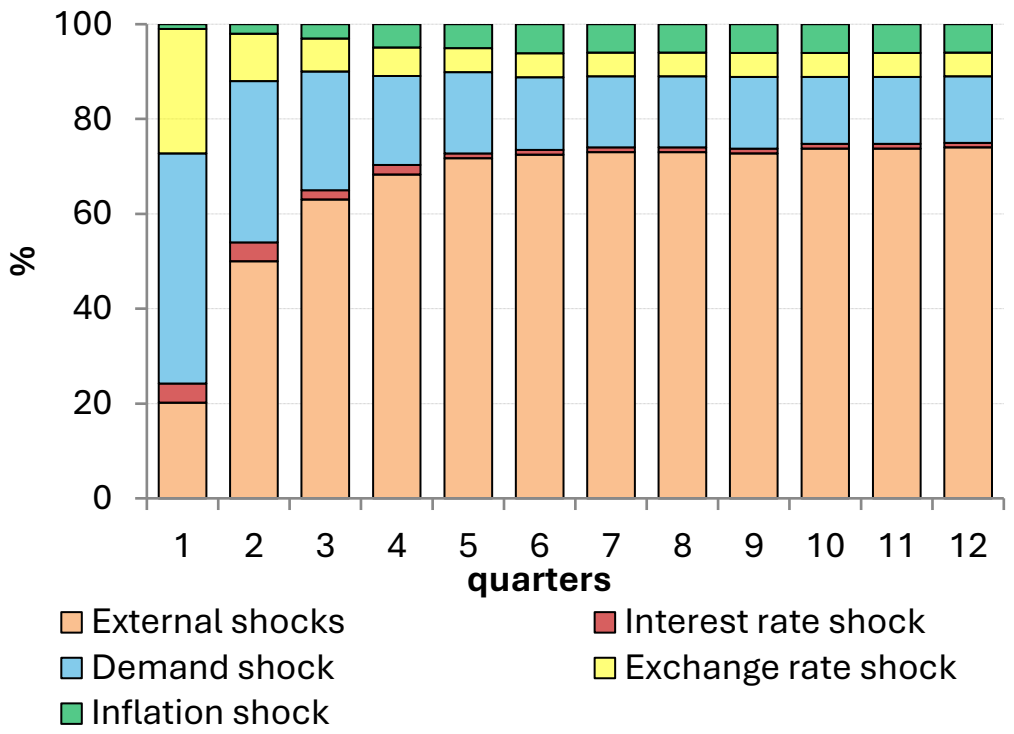
Source: Czech National Bank. Monetary Policy Report – Spring 2026

Variance decomposition: splitting the uncertainty

- At each horizon, the forecast-error variance is sliced into the part coming from each structural shock.
- The slices always **add up to 100%**.
- Short run: mostly the variable's **own shock**.
  Longer run: the other shocks grow.

It's the same Θ-matrices as the IRF – now squared and summed.

Output gap forecast error variance decomposition



Source: Kharitonchik, A., Dmitriev, D. (2018) A study of the monetary policy transmission mechanism in the Republic of Belarus. *Bank Bulletin Journal. Thematic Issue "Bank's Research No 13"*.

IRF vs. error variance decomposition

Impulse Response

What is the effect?

- Direction & size of a shock's effect
- How the effect evolves over time
- **“Reaction”**

Variance Decomposition

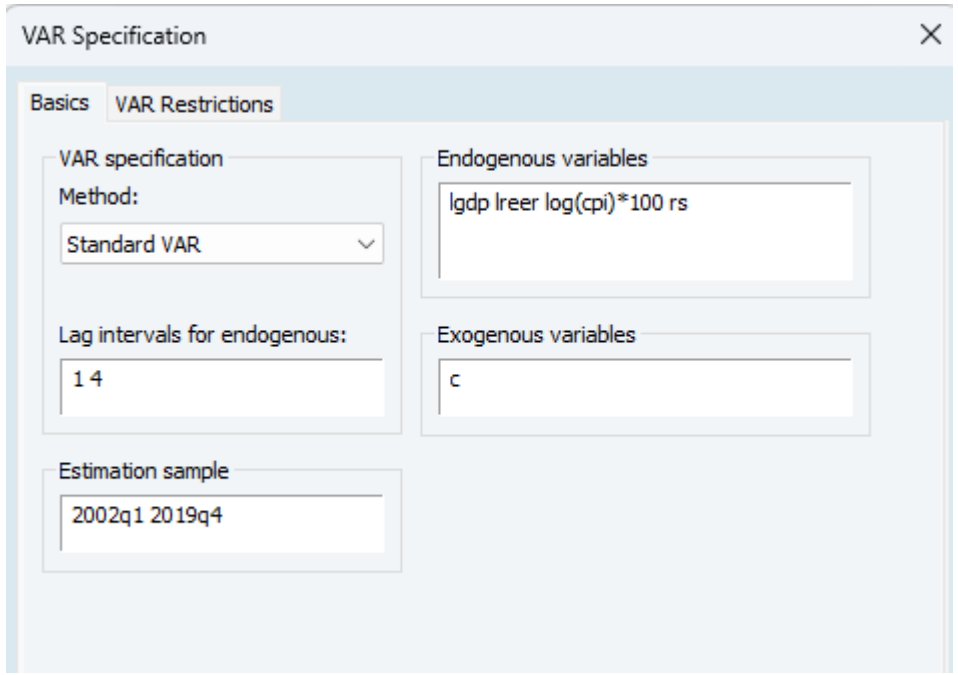
How much does it matter?

- Relative importance of shocks
- Shows sources of fluctuations
- **“Contribution”**

Both are built from the same identified shocks. **No identification → no meaningful IRF or decomposition.**

Impulse response analysis in EViews (1)

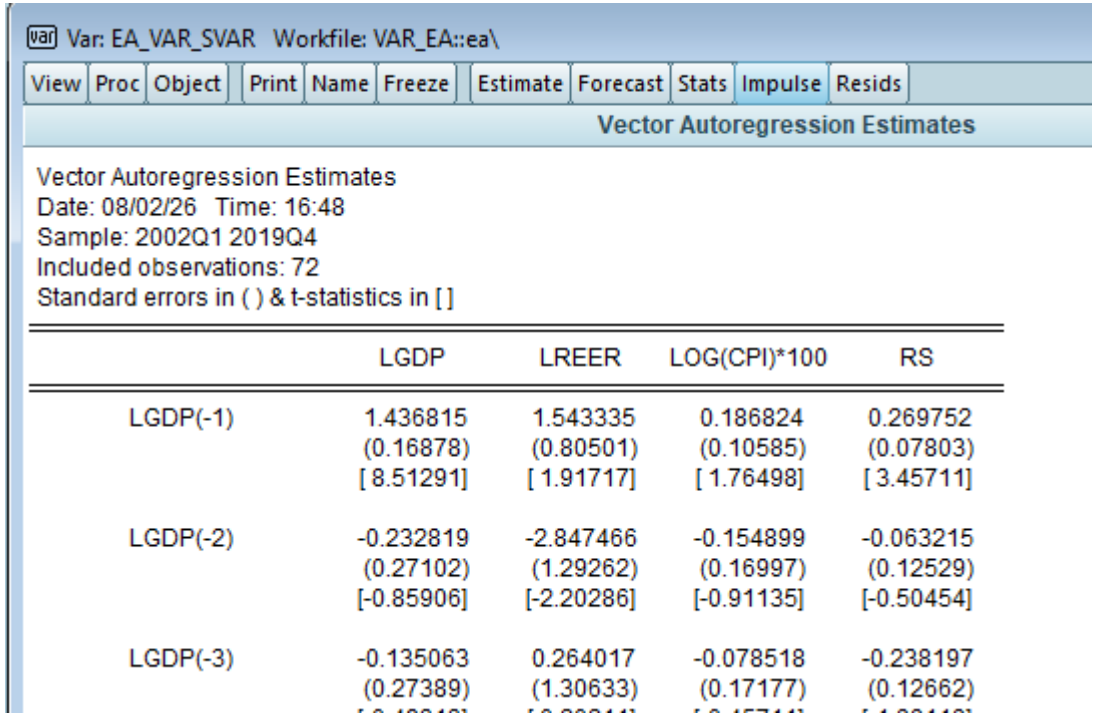
Let’s re-estimate the EA reduced-form VAR using levels of macroeconomic variables:



Vector Autoregression Estimates
Date: 08/02/26 Time: 16:51
Sample: 2002Q1 2019Q4
Included observations: 72
Standard errors in ( ) & t-statistics in [ ]

	LGDP	LREER	LOG(CPI)*100	RS
LGDP(-1)	1.436815	1.543335	0.186824	0.269752
	(0.16878)	(0.80501)	(0.10585)	(0.07803)
	[8.51291]	[1.91717]	[1.76498]	[3.45711]
LGDP(-2)	-0.232819	-2.847466	-0.154899	-0.063215
	(0.27102)	(1.29262)	(0.16997)	(0.12529)
	[-0.85906]	[-2.20286]	[-0.91135]	[-0.50454]
LGDP(-3)	-0.135063	0.264017	-0.078518	-0.238197
	(0.27389)	(1.30633)	(0.17177)	(0.12662)
	[-0.49313]	[0.20211]	[-0.45711]	[-1.88118]
LGDP(-4)	-0.059911	1.160332	0.074443	0.013936
	(0.19437)	(0.92704)	(0.12190)	(0.08986)
	[-0.30824]	[1.25166]	[0.61071]	[0.15509]
LREER(-1)	0.048192	1.175660	-0.003068	0.022632
	(0.02827)	(0.13484)	(0.01773)	(0.01307)
	[1.70465]	[8.71899]	[-0.17306]	[1.73165]
LREER(-2)	-0.037424	-0.459616	0.016243	-0.010441
	(0.04466)	(0.21301)	(0.02801)	(0.02065)
	[-0.83795]	[-2.15767]	[0.57992]	[-0.50568]
LREER(-3)	-0.031098	0.159080	-0.024303	-0.030146
	(0.04347)	(0.20731)	(0.02726)	(0.02009)
	[-0.71545]	[0.76735]	[-0.89154]	[-1.50024]
LREER(-4)	0.033558	-0.044504	0.014306	0.024519
	(0.02593)	(0.12368)	(0.01626)	(0.01199)
	[1.29418]	[-0.35984]	[0.87970]	[2.04538]
LOG(CPI(-1))*100	-0.124163	0.148714	1.351301	0.254644
	(0.21899)	(1.04450)	(0.13734)	(0.10124)
	[-0.56697]	[0.14238]	[9.83897]	[2.51520]
LOG(CPI(-2))*100	-0.253194	0.171265	-0.346408	-0.304607
	(0.36841)	(1.75715)	(0.23105)	(0.17032)
	[-0.68726]	[0.09747]	[-1.49930]	[-1.78846]
LOG(CPI(-3))*100	0.239571	-1.054546	0.091050	0.130589
	(0.37318)	(1.77990)	(0.23404)	(0.17252)
	[0.64197]	[-0.59247]	[0.38904]	[0.75694]
LOG(CPI(-4))*100	0.094951	0.607940	-0.128175	-0.068640
	(0.23531)	(1.12233)	(0.14758)	(0.10879)
	[0.40351]	[0.54168]	[-0.86854]	[-0.63096]
RS(-1)	-0.039047	-2.040265	-0.199893	0.984112
	(0.35455)	(1.69105)	(0.22236)	(0.16391)
	[-0.11013]	[-1.20651]	[-0.89898]	[6.00395]
RS(-2)	-0.652056	6.028870	0.229157	-0.275841
	(0.51439)	(2.45341)	(0.32260)	(0.23781)
	[-1.26763]	[2.45735]	[0.71035]	[-1.15995]
RS(-3)	0.806727	-3.667255	-0.032361	0.329081
	(0.49402)	(2.35627)	(0.30983)	(0.22839)
	[1.63297]	[-1.55638]	[-0.10445]	[1.44087]
RS(-4)	-0.341835	0.067611	-0.047815	-0.085002
	(0.30327)	(1.44647)	(0.19020)	(0.14020)
	[-1.12716]	[0.04674]	[-0.25140]	[-0.60627]
C	-0.622631	-45.18271	-28.75271	17.80900
	(35.6502)	(170.036)	(22.3580)	(16.4813)
	[-0.01746]	[-0.26572]	[-1.28602]	[1.08056]
R-squared	0.996553	0.922236	0.999264	0.987087
Adj. R-squared	0.995550	0.899614	0.999050	0.983331
Sum sq. resids	9.758013	221.9815	3.837969	2.085547
S.E. equation	0.421211	2.008986	0.264161	0.194728
F-statistic	993.6947	40.76691	4669.105	262.7750
Log likelihood	-30.21479	-142.6970	3.378449	25.33528
Akaike AIC	1.311522	4.436027	0.378376	-0.231535
Schwarz SC	1.849068	4.973573	0.915923	0.306011
Mean dependent	1482.321	465.3287	428.1734	1.093379
S.D. dependent	6.314029	6.340750	8.571894	1.508251

Press “Impulse”



Impulse response analysis in EViews (2)

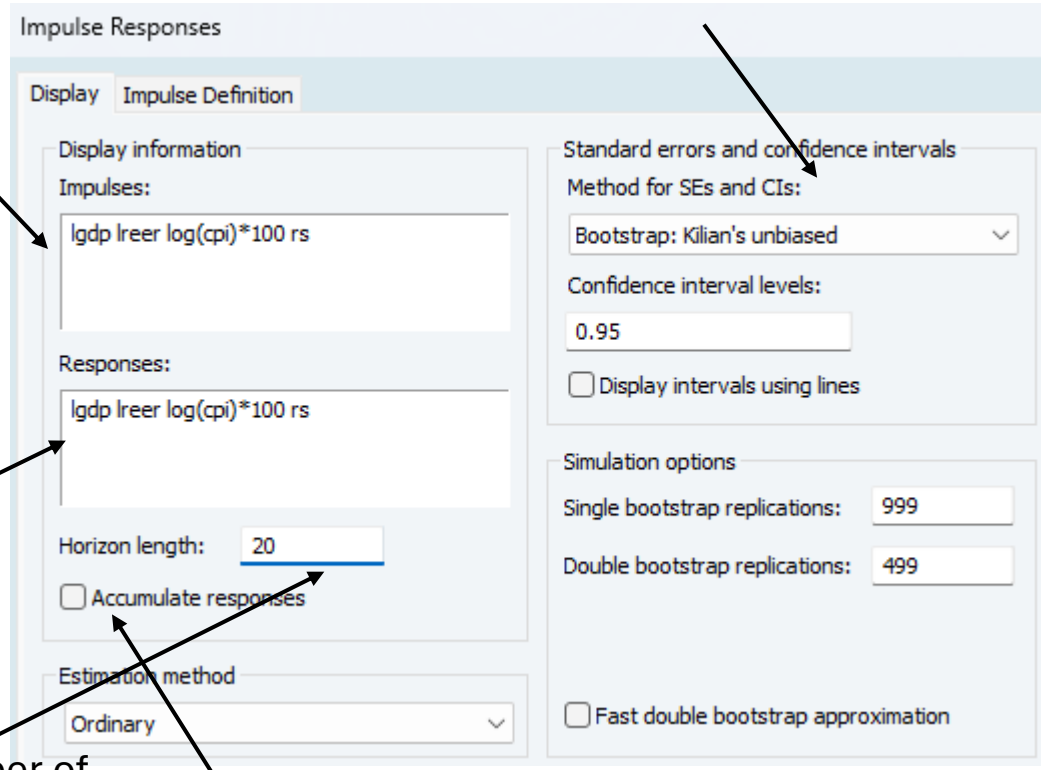
Specify how to compute bands (bootstrapped suggested for macro data)

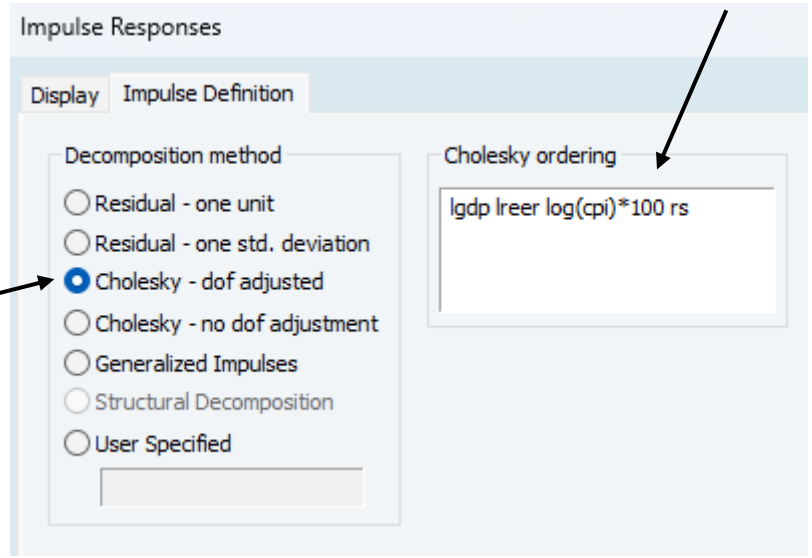
Specify the order: from the "least" endogenous to the "most" endogenous

Specify shocks to analyse

Specify responses to analyse

Recursive ordering

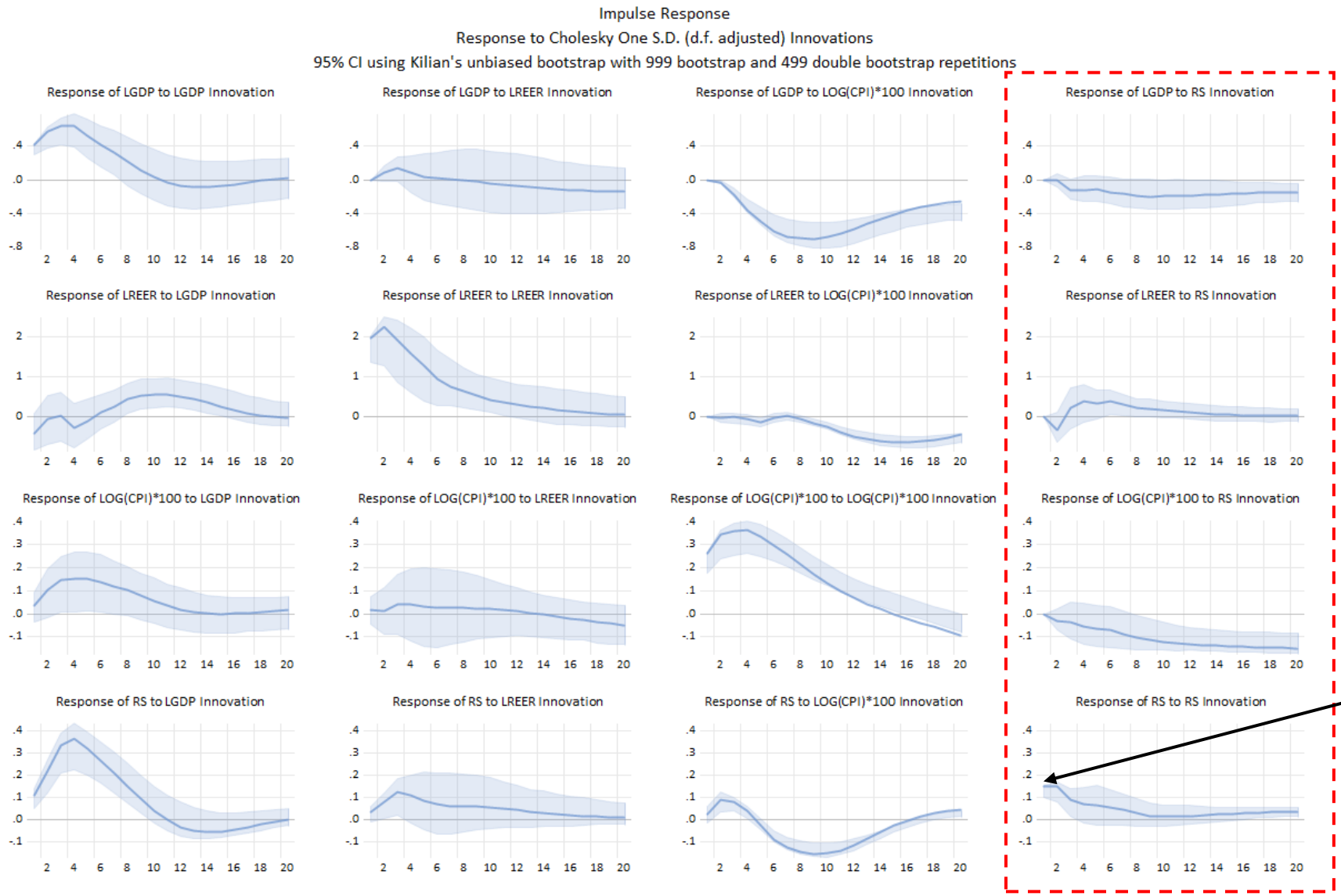




Type number of periods after the shock to analyses

Use accumulated responses if you want to analyze the cumulative response (for example, if VAR is in differences, then the cumulative response will show the change in the variable level). Not relevant for VAR in levels.

Impulse response analysis in EViews (3)

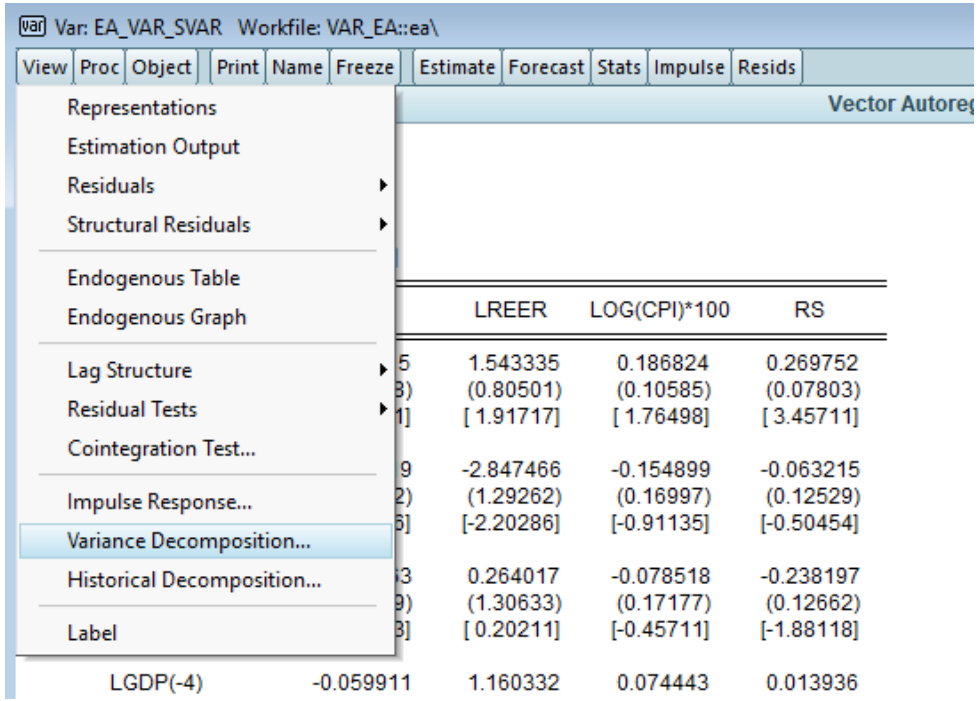


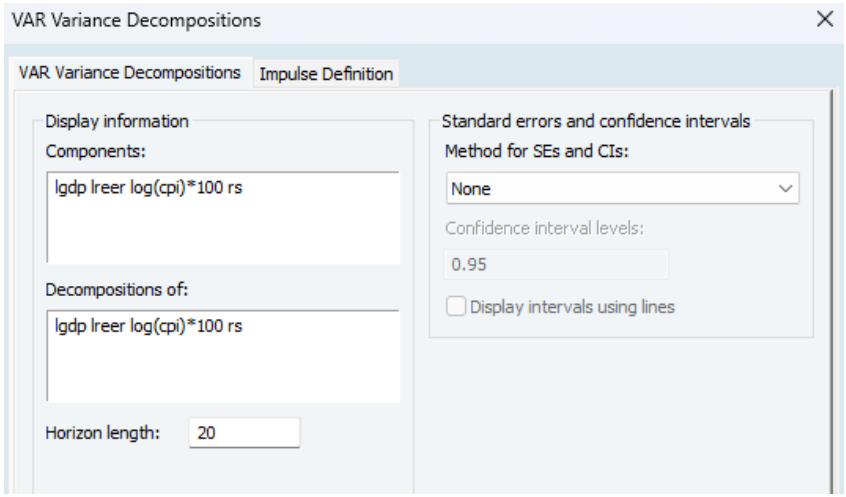
A contractionary monetary-policy shock in this model raises the rate, cools output, lowers inflation, and briefly appreciates the currency.

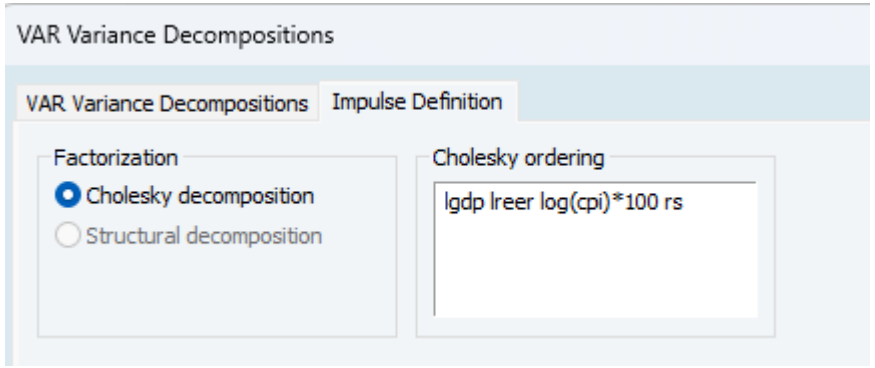
But read each claim against its band, since output and the exchange-rate effects are only marginally distinguishable from zero.

Dividing the RS column by its impact size (≈0.15) rescales the responses to a one-percentage-point monetary-policy shock.

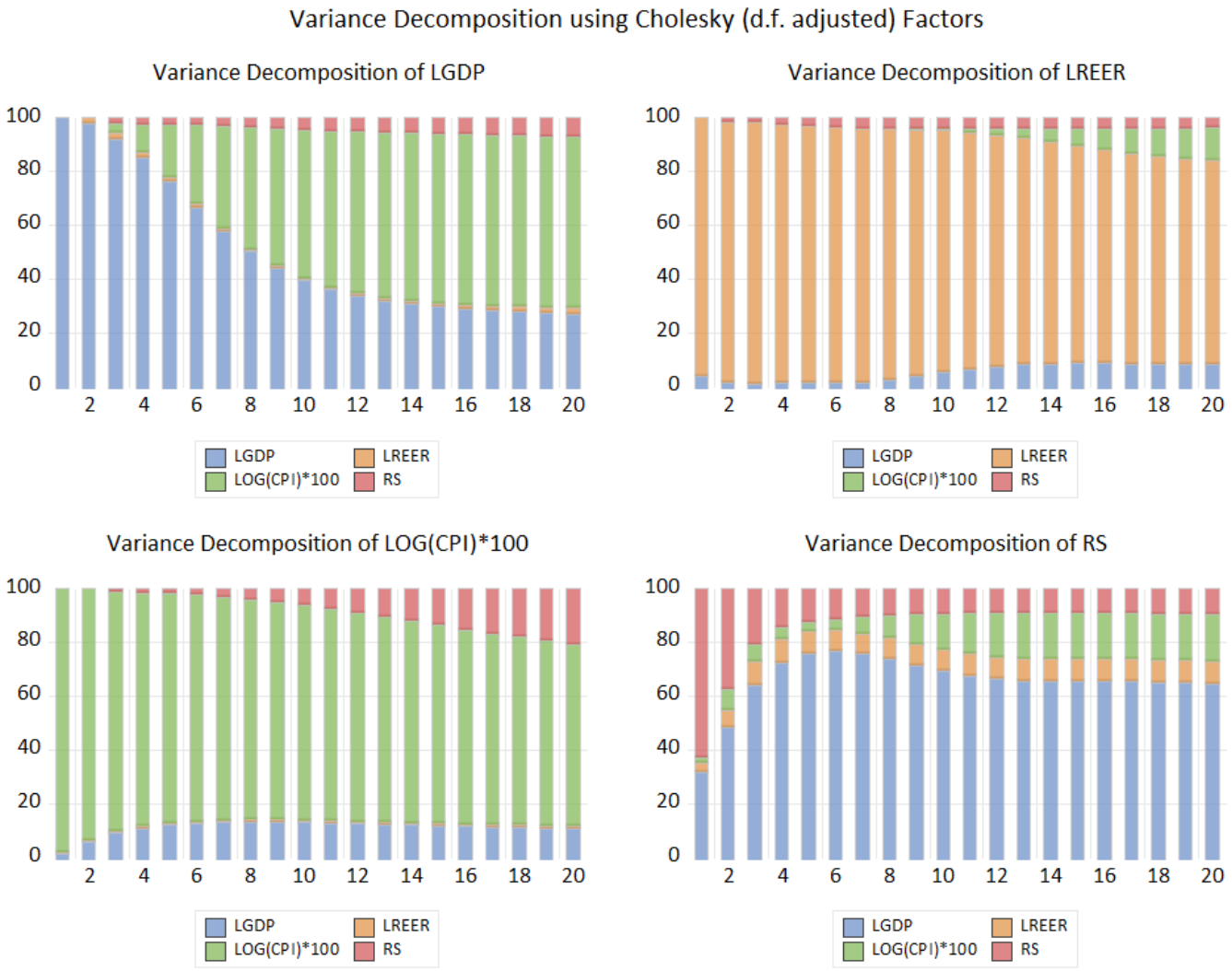
Error variance decomposition in EViews (1)







Error variance decomposition in EViews (2)



Long-horizon forecast errors are driven mostly by *other* shocks:

- output uncertainty is dominated by price (supply) shocks (~65%);
- the policy rate's own forecast errors are driven largely by output shocks (~65%): monetary policy is mostly a systematic reaction to the cycle. Genuine policy surprises explain only a small and slowly-growing share of inflation's forecast variance (~20%).

Why do we need VARs? Once again

	Reduced-form VAR	Structural VAR
Question	What comes next?	What caused what?
Needs	Estimation only	Estimation and identification
Output	Forecasts, scenarios	Impulse responses, variance decompositions